

The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

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Abstract

Whether expert information about occupational automation risk increases demand for protective government policy remains an open question. In this paper, we propose and test a conditional model of policy demand to expert risk signals depending on two conditions: trust in the expert source, which determines whether the signal is absorbed at all, and individual disposition, which requires a person to be specially sensitive to the communicated risk. The model is examined in a pre-registered between-subjects experiment with 901 United States workers (475 treated, 426 control), in which a three-step protocol explains how experts assess automation risk, elicits each respondent's belief about the relative risk of their own occupation, and then reveals the corresponding expert estimate, so that the intensity of the treatment varies with each respondent's occupation. Policy demand is measured by a five-item battery spanning unemployment benefits, federal regulation of AI deployment, retraining programs, labor protections, and limits on how many jobs AI may automate, scored with a graded response model. Because trust in the source is observed only among treated respondents, stratum-specific effects are recovered by predicting the trust rating from fifteen pre-treatment covariates, converting that prediction into a principal score, and applying Hájek weights to both arms; the resulting estimands are identified under randomization together with principal ignorability. The unconditional effect of the treatment on policy demand is positive but modest, ranging from 0.22 to 0.28 standard deviations across covariate specifications. It is estimated at 0.40 standard deviations [0.22, 0.61] among respondents whose trust in the source is higher and at 0.14 [-0.02, 0.27] among those whose trust is lower, and it reaches 0.69 [0.36, 0.98] among higher-trust respondents located in a pre-specified corner defined by the top tercile of both trait anxiety and own-occupation automation risk, each measured before treatment, against 0.29 outside that corner; the corresponding increment among lower-trust respondents is indistinguishable from zero. A sensitivity analysis places the breakdown point of the higher-trust estimate at a hidden differential treatment response of -1.17 standard deviations, a magnitude opposite in sign to the mechanism under study. Reported minimum detectable effects indicate that the higher-trust quantities are well powered, whereas the difference between strata and the corner increments are not. The results are consistent with an interpretation of weak aggregate estimates in this literature as composition across heterogeneous response pathways rather than as stability of policy preferences.

1 Automation Risk, Economic Insecurity, and Demand for Government Protection

Classical models in political economy suggest that citizens respond to new economic risks by updating their policy preferences in favor of government protection and redistribution (Iversen and Soskice, 2001; Rehm, 2009; Margalit, 2019). Yet, empirical evidence from survey experiments often finds a disconnect between increased perceptions of economic vulnerability and changes in policy preferences. The rapid growth of Artificial Intelligence (AI) and its looming potential to automate jobs provide a test case to understand this puzzle. For instance, Zhang (2022) conducted three survey experiments on a US sample: in two of the three, informational treatments about AI automation risks increased perceptions of economic vulnerability, yet policy preferences

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remained unchanged across all three. Further, Magistro et al. (2024) varied the frame content of automation information (cost-only versus negative and positive trade-offs frames), measuring policy items contextualized as responses to the automation shock. Nevertheless, across respondents from four OECD countries, they found no differences in policy preferences between treatment groups.

Yet, the pattern is not uniformly null. Other studies find positive effects of automation risk on support for government intervention, though the direction of policy demand varies and it is often conditional: Gallego et al. (2022) find that concerned workers favor protectionist measures (slowing down technology) rather than redistribution, Jeffrey (2021) shows that fairness rhetoric is needed to produce gains in redistributive support, concentrated among bachelor's-educated respondents; and Heinrich and Witko (2025) find that AI labor-market information shifts AI-regulation support in opposite directions across AI-knowledge subgroups, with high-knowledge respondents demanding more regulation and low-knowledge respondents less.

A common feature of these studies is that they often rely on information treatments that are not specifically anchored to respondents' own occupational automation risk. For example, Borwein et al. (2024) treatment describes the general risk of AI automation without explicitly connecting it to respondents' own job or industry. Other studies use specific other-regarding frames in which respondents read about layoffs at a toymaker (Wu, 2023), manufacturing firm (Magistro et al., 2024), or food manufacturer (Ladreit, 2022), none of which is matched to the respondent's own industry, occupational family, or specific job. These designs often highlight the sociotropic effects of automation risk, which may not be the most direct way to capture self-interest motivations as a driver of policy demand.

Only a couple studies depart from this pattern by delivering automation-risk information keyed to the respondent's own occupation. Golin and Rauh (2023) pipe in each respondent's SOC 2018 narrow occupation category and present a job-loss probability framed as either an expert estimate or as an aggregated belief of similarly-employed workers.¹ Zhang (2022) elicits the respondent's job from a 214-item dropdown and shows the corresponding Frey-Osborne or Nedelkoska-Quintini automation probability. These two designs are the closest to frame information in a way that is grounded to a respondent actual automation risk.

Current gaps are not only design-based. Across these studies, what is also absent is a theoretical account of how citizens engage automation-risk information before policy demand can update. In this paper, we argue that citizens do not absorb risk signals as neutral content. When the signal runs against prior beliefs about their own work and risk, absorbing it is uncomfortable; respondents may resolve the tension by questioning the credibility of the source rather than incorporating the message. And when respondents already believe themselves to be at risk on the basis of signals from their own informational environment, there is little room left to update further; they are, in a sense, pre-treated. These two channels, whether the signal is accepted and how much room there is to update on it, have been largely unaddressed in this literature.

I test this theoretical framework, termed the "Conditional Activation Model (CAM)," in a novel pre-registered survey experiment with N=901 U.S. workers. The treatment procedure involves three informational steps: first, an excerpt briefly explains how experts assess occupational automation risk by looking at the composition of tasks of a given job; second, I measure respondents' perception of their relative automation risk; and, finally, a passage contrasts reported subjective beliefs with the expert estimates. A pure control group skips the informational procedure.

Policy demand is then measured as the support for five interventions targeting the workforce impact of AI-driven automation: expanding unemployment benefits, enacting AI regulations, providing retraining funds, strengthening labor protections, and limiting automation. Comparing the treatment and control groups, I find an unconditional treatment effect that is positive but modest. Consistent with the CAM framework, a larger effect concentrates among respondents who trust the source and are disposed to act on the risk it communicates. For those who trust the source, the conditional average treatment effect reaches approximately 0.40 standard de-

¹However, the displayed numbers in both arms come from aggregated self-reports collected as part of the Covid Inequality Project (May 2020 wave). The authors disclose that "the flattering addition of the term 'experts' is for polarizing expositional purposes" (Golin and Rauh, 2023, p. 4, fn. 5). This design limits the ecological validity of the findings.

viations on the latent preference for government intervention ($d = 0.402$, 95% CI [0.221, 0.608]). Finally, this effect reaches 0.69 standard deviations among those who trust the source and whose trait anxiety and perceived automation risk are both high (DeYoung, Quilty and Peterson, 2007), a stable personality trait of threat vigilance ($d = 0.685$, 95% CI [0.364, 0.980]). These highly vigilant individuals are theorized to be oriented toward accuracy-based reasoning, which may make them more responsive to credible risk signals that run against their prior beliefs.

2 Theory: The Conditional Activation Model

This section develops the Conditional Activation Model (CAM), which holds that expert information about automation risk shapes policy demand only through two sequential conditions. The first is whether the respondent accepts the source of the information as credible; when trust is absent, the message is discounted regardless of content. The second is whether accepted information is intense enough, and the respondent disposed enough, to translate into policy demand: the level of automation risk communicated and the respondent's trait anxiety jointly determine whether the signal is elaborated on rather than set aside.

Source credibility functions as a cognitive shortcut: when individuals trust a message's source, they are more likely to accept its content, and when they distrust it, the same information is discounted or rejected (Lupia and McCubbins, 1998; Zaller, 1992). Source credibility moderates framing effects (Druckman, 2001), and trust in scientific experts carries systematic partisan asymmetries (Gauchat, 2012; Motta, 2018). If trust conditions how expert information is processed, the null aggregate effects reported by several automation experiments may reflect unmodeled heterogeneity in source acceptance rather than genuine preference stability: respondents update beliefs about the future of work without moving policy preferences (Zhang, 2022), remain unmoved even when policy items are explicitly automation-specific (Magistro et al., 2024), and register heightened worry with no downstream policy shift (Ladreit, 2022). Heinrich and Witko (2025) find policy effects concentrated among respondents with high prior technology knowledge, a construct plausibly correlated with trust in expert forecasts.

Acceptance of the source is necessary but not sufficient. Affective Intelligence Theory holds that anxiety functions as a dispositional trigger for vigilance: individuals higher in trait anxiety are more inclined toward accuracy-motivated, deliberative engagement with threatening information rather than habitual reliance on standing dispositions (Marcus, Neuman and MacKuen, 2000; Valentino et al., 2008), and Albertson and Gadarian (2015) show that anxious citizens lean more, not less, on trusted experts, linking this disposition back to the trust gate. CAM predicts that, among respondents who trust the source, demand for protective policy should rise with the level of automation risk communicated, and that this relationship should be strongest among respondents higher in trait anxiety: greater perceived risk and greater dispositional vigilance jointly raise the likelihood that a signal is acted on rather than set aside.

Trait anxiety denotes a stable disposition to experience anxiety across situations. State anxiety denotes the transient reaction to an immediate situation. The State-Trait Anxiety Inventory, the canonical measure, separates the two (Spielberger et al., 1983). Its trait items describe standing tendencies, such as "I feel nervous and restless" and "I worry too much over something that really does not matter"; its state items describe the present moment, such as "I am tense" and "I am worried." The disposition has documented economic correlates; trait-anxious individuals choose more conservative investments (Gambetti and Giusberti, 2012). The distinction carries the design logic of the second gate. A stable disposition can moderate how a worker processes a risk signal; a transient state can be produced by the signal itself. CAM's moderator is the disposition, measured before treatment.

This structure has a direct implication for what an experiment should find. The unconditional average treatment effect pools three groups the model distinguishes: respondents who discount the source, whose response should be attenuated or absent; respondents who accept the source but face modest communicated risk, or lack the disposition to dwell on it, whose response should be weak; and respondents who accept the source, face high communicated risk, and are high in trait anxiety, whose response should be largest. Averaging over the three

dilutes the strongest response.² A modest aggregate effect is therefore consistent with the model, and it is uninformative on its own: the same average is compatible with a uniform weak response across the sample and with a strong response concentrated in a narrow cell. Arntz, Blesse and Doerrenberg (2024) document a version of this pattern directly, finding opposing treatment effects conditional on prior beliefs that cancel in aggregate. The content of the model lies in its conditional predictions, which specify where the response should concentrate.

The model yields four predictions, ordered from the aggregate to the joint condition.

H1 (Modest or Null Aggregate Effect). Because the average treatment effect pools the responsive cell with weaker responses elsewhere, and the shares of the three groups in the population are not known in advance, the aggregate effect should be modest at most and may be indistinguishable from zero. A large aggregate effect would suggest the conditional structure matters less than the model holds; a null one would not, on its own, separate the model from uniform unresponsiveness. That separation falls to the conditional predictions.

H2 (Trust Gradient). Treatment effects should be larger among respondents who trust the source than among those who do not. Comparable effects across trust levels would indicate that source acceptance does not condition processing in this domain.

H3 (Risk-Level Gradient). Among respondents who trust the source, the effect should rise with the level of communicated automation risk. A flat profile across risk levels would indicate that acceptance alone, rather than the content of what is accepted, drives the response.

H4 (Anxiety Amplification). The risk gradient should be steepest among respondents high in trait anxiety, so the largest effect should sit where trust, high communicated risk, and high trait anxiety coincide. An equally strong response among low-anxiety respondents who trust the source would indicate that disposition adds nothing beyond acceptance and content.

These predictions concern whether policy demand activates, not which policy gains support; the direction demand takes once activated is shaped by prior political commitments, and the battery used here emphasizes government-led responses.

3 Research Design and Methods

3.1 Experimental Design and Sample

We conducted a pre-registered, between-subjects randomized experiment with U.S. workers recruited via CloudResearch Connect. Eligibility required participants to be ages 18–65, currently in the active labor force (employed at least part-time, or unemployed and actively seeking work), and English-speaking. After applying data-preparation filters, the analytic sample comprised $N = 901$ respondents, randomly assigned to either the treatment condition (receiving personalized automation risk information) or a pure control condition.

3.2 Treatment: The Three-Step Belief Correction Procedure

The treatment is designed to expose each respondent to an expert assessment of their own occupation’s automation risk, and to measure their belief update potential. To achieve this, we implement a three-step procedure that informs respondents on how experts assess automation risk in general, elicits the respondent’s prior belief about their occupation’s risk relative to others, and then shows the expert estimate alongside the stated prior belief. The sequencing produces a *Belief Gap*, defined as the difference between each the expert assessment of relative automation risk and each respondent’s prior.

Step 1: Expert Context. All treatment-group respondents first read a neutral primer on how experts assess automation risk. The passage describes the general logic of expert assessment: experts examine the component

²Druckman and Leeper (2012) document this dilution in framing experiments: respondents with strong prior attitudes resist the message, and the pooled average understates the effect among the movable subgroup. The source of resistance differs here, source distrust rather than prior exposure, but the arithmetic of pooling is the same.

tasks of each occupation, identify which tasks are routine and clearly defined (and thus susceptible to automation), and note that jobs with more data-intensive, online, or repetitive workflows face higher risk. This initial explanation does not yet reveal the specific source (Frey and Osborne, 2017) or the respondent's personal score, it merely educates respondents about the conceptual basis for expert assessments.

Step 2: Subjective Belief Elicitation. Before revealing any personalized information, we ask respondents to state their own belief about where their job falls on the automation risk distribution (0 = safest; 100 = riskiest). The wording is deliberately relative³: respondents are asked to place their occupation on a single ranked line of all U.S. workers, from least to most at risk of automation, which captures a comparative belief without requiring respondents to interpret the underlying scale as a percentile. This elicitation occurs *after* respondents understand the methodology but *before* they see expert data about their occupation. This sequencing creates psychological commitment to a prior estimate: respondents anchor on a self-generated belief, which amplifies the salience of the Belief Gap revealed in the next step.

Step 3: Personalized Correction. Finally, we contrast the expert risk score for the respondent's specific occupation with their own estimate. Along with contrasting the two estimates, we faithfully attribute the source to Oxford researchers Dr. Carl Benedikt Frey and Dr. Michael A. Osborne, who developed the original occupation-level automation risk probabilities. The score respondents see is not the raw Frey-Osborne probability but an employment-weighted percentile rank within the U.S. civilian labor force: for each occupation, it reports the share of workers in occupations with the same or lower Frey-Osborne probability.⁴ A short excerpt places the respondent's stated belief and the expert estimate on the same 0–100 scale, making the Belief Gap (expert minus own relative risk) visible at the individual level. The gap this step reveals is large: prior beliefs are nearly unrelated to the expert percentile, and their accuracy is unrelated to respondents' declared certainty (SI Section S7.1).

Control Group. Respondents assigned to control proceed directly to outcome measures without receiving any automation-related information, belief elicitation, or expert estimates.

3.3 Measurement

Primary Outcome: Policy Demand Index. The primary outcome is a five-item index measuring support for government policy responses to AI-driven automation. Respondents rated each item on a 7-point scale (1 = Strongly Oppose; 7 = Strongly Support):

1. *Expanding unemployment benefits and income subsidies.*
2. *Introducing stricter federal regulations on AI deployment, such as requiring companies to undergo ethical AI reviews.*
3. *Funding government or private-sector retraining programs to help workers transition to new careers.*
4. *Strengthening labor protections, such as supporting unionization efforts in AI-exposed industries.*
5. *Limiting the number of jobs that AI can automate within companies to protect human employment.*

Trust in Information Source. Trust in the expert risk estimate is measured via a single item presented only to the treatment group⁵, immediately after viewing the Frey-Osborne risk score for their occupation:

³We elicit subjective beliefs in relative terms (rank in the workforce) rather than as absolute automation probabilities for two reasons. First, relative risk is plausibly a more faithful representation of how individuals subjectively reason about job security, since people typically frame their position by comparison to peers and reference groups rather than as point probabilities; a rank-based elicitation captures this directly. Second, in labor-market contexts, relative position is arguably the more decision-relevant quantity for choices about reskilling, mobility, and bargaining: absolute automation probabilities are highly uncertain and depend on a forward-looking time horizon whose definition shifts as AI capabilities evolve, whereas relative position is a more stable signal of an occupation's exposure.

⁴We compute the percentile via an employment-weighted empirical CDF over Frey-Osborne probabilities, weighting each occupation by its civilian labor force count (employed plus unemployed, ages 16 and over) from the BLS Current Population Survey Basic Monthly microdata. Frey-Osborne occupation codes (SOC 2010) are crosswalked to SOC 2018 to align with the CPS coding.

⁵This item is post-treatment by design: trust in expert estimates cannot be elicited without first revealing those estimates. The post-treatment and treatment-group-only nature of this measure creates identification challenges (endogeneity, missing data for controls) that are addressed in Sections 3.4 and 3.5.

“To what extent do you think the Oxford researchers are a trustworthy source of information about job automation risk?” (1 = Not at all trustworthy; 5 = Extremely trustworthy)

Trait Anxiety. Trait anxiety is measured pre-treatment for all respondents using a four-item battery drawn from the Big Five Aspects Scale (DeYoung, Quilty and Peterson, 2007), targeting the anxiety facet of neuroticism as classified by Johnson (2014) and Maples et al. (2014): “I see myself as anxious, easily upset,” “I feel threatened easily,” “I worry about things,” and “I am afraid of many things” (7-point agreement scale).

Intermediate Outcomes. Two post-treatment batteries capture state-dependent economic perceptions: a four-item Job Insecurity and Automation Beliefs scale (7-point agreement) and a seven-item Perceived Impacts of AI scale (5-point likelihood). Full item wordings appear in SI Section S8.

Covariates. Pre-treatment covariates entering the outcome model include political ideology (7-point liberal-conservative scale), party identification, education, numeracy (objective math task), religiosity, locus of control, and demographics (age, gender, race/ethnicity). The trust measurement equation includes a separate set of covariates: prior exposure to automation risk information, constructed as a five-level ordinal index from a two-step item sequence (whether respondents have previously encountered information about AI or automation affecting jobs like theirs, and if so, how much), along with the Belief Gap and its square.

3.4 Estimation Strategy and Identification

H2 compares treatment effects across levels of trust in the source. Two features of the design stand in the way of that comparison. Trust in the expert estimate was measured after the treatment delivered that estimate, and only in the treated arm: control respondents never saw the estimate, so no rating exists for them. The control arm therefore cannot be split on trust. The treated arm can be, but the split is invalid: high-trust treated respondents are a subgroup selected by a post-treatment variable, and comparing them with the full control group mixes the treatment effect with whatever sorts respondents into high trust.

The estimator instead uses the part of trust that is predictable before treatment. Trust ratings vary systematically with stable pre-treatment characteristics, and a model fit in the treated arm, where the rating is observed, recovers this predictable component. Because every input is measured before treatment, the model assigns each respondent in both arms a probability of belonging to the high-trust stratum. Weighting each arm by that probability, treated arm included, compares two groups built identically from information the treatment could not have touched. The observed rating never forms a comparison group. It only teaches the model, in the one arm where it exists, how trust varies with pre-treatment characteristics.

Respondents are indexed by i , with $N = 901$. The assignment indicator D_i equals 1 for the 475 treated respondents and 0 for the 426 controls. Policy demand η_i is the standardized score from the measurement model above. Its potential values are $\eta_i(1)$ with treatment and $\eta_i(0)$ without, and the observed score is $\eta_i = \eta_i(D_i)$. The trust rating Q_i is the five-point rating a respondent gives the expert estimate when treated, so it is observed only in the treated arm. The stratum indicator $S_i = \mathbf{1}\{Q_i \geq 4\}$ marks the high-trust stratum. It is defined for every respondent through the rating they would give if treated, and it is observed only where that rating is.

The targets are the two stratum treatment effects

$$\tau_{\text{high}} = \mathbb{E}[\eta(1) - \eta(0) \mid S = 1], \quad \tau_{\text{low}} = \mathbb{E}[\eta(1) - \eta(0) \mid S = 0]. \quad (1)$$

The trust gradient of H2 is $\tau_{\text{high}} - \tau_{\text{low}}$. The average treatment effect is the mixture $\Pr(S = 1) \tau_{\text{high}} + \Pr(S = 0) \tau_{\text{low}}$.

The strategy reaches these targets in three steps.

- A random forest predicts the trust rating Q_i from pre-treatment characteristics. It is trained in the treated arm, where the rating exists, and computed for every respondent.
- A logistic regression converts the prediction into $p_i = \Pr(S_i = 1 \mid \hat{Q}_i)$, each respondent’s probability of belonging to the high-trust stratum, with $\hat{Q}_i$ the standardized prediction.

- The outcomes in each arm are averaged with weights p_i and the weighted means are differenced, which estimates τ_{high} . The same construction with weights $1 - p_i$ estimates τ_{low} . This is the **Hájek estimator** (Hájek, 1971), the self-normalized variant of Horvitz–Thompson inverse-probability weighting (Horvitz and Thompson, 1952), with principal scores as the weights: each arm’s mean is normalized by the realized sum of the weights rather than by the sample size, so the weights average one in the sample.

Subsection 3.5 specifies each step.

Two assumptions identify the trust-stratum effects. Randomization balances every pre-treatment characteristic across arms. The prediction model reads only those characteristics, so its fitted probability describes treated and control respondents on equal terms. Randomization says nothing about the strata themselves. That is the role of principal ignorability (Frangakis and Rubin, 2002; Ding and Lu, 2017). The assumption states that among respondents with the same predicted trust, membership in the high-trust stratum adds no information about the policy demand they would express without treatment. Randomization holds by design. Principal ignorability must be assumed. The rating that would test it directly does not exist in the control arm. The estimates are identified under the assumption. Its plausibility can still be assessed. A sensitivity analysis measures how far the assumption must fail before each estimate reaches zero. Section 4 reports these breakdown values.

The weighting yields three quantities. Each arm is weighted by the fitted probability, and the difference between the two weighted means is the treatment effect in the high-trust stratum. The same construction with the complementary probability gives the effect in the lower-trust stratum. The difference between the two stratum effects is the trust gradient that H2 predicts. The unweighted average treatment effect mixes the two stratum effects in proportion to the stratum shares. A modest average is therefore compatible with a strong response inside the high-trust stratum. This is the composition argument of Section 2 in estimator form.

SI Section S1 states the assumptions formally, derives the breakdown value the sensitivity analysis reports, and situates the construction relative to principal-score methods for observed post-treatment variables.

3.5 Trust-Stratum Estimation

The estimator proceeds in three steps. First, a random forest (500 trees, maximum depth 10, minimum leaf size 5) predicts the five-point trust rating from fifteen pre-treatment terms.⁶ The forest is trained on the treated arm, where the rating exists, and its prediction is computed for every respondent and standardized over the full sample. Second, a logistic regression converts this prediction into each respondent’s probability of belonging to the high-trust stratum. High trust means a rating of 4 or 5 on the five-point scale, the top two categories. The regression is fit in the treated arm, where the indicator is observed, and the fitted probability is computed for every respondent. This is the fitted probability of the previous subsection. Third, the outcomes in each arm are averaged with each respondent weighted by that probability, and the difference between the two weighted averages is the treatment effect in the high-trust stratum.⁷ The outcome throughout is the policy-demand score described above.

Standard errors are computed by bootstrap with 2,000 replicates. Each replicate resamples respondents within their arm. The random forest, the logistic regression, and the weights are refit from the resampled data. The intervals therefore carry the uncertainty of the prediction model along with sampling variation. A resampled treated arm occasionally yields a degenerate prediction or contains only one stratum. Such replicates are dropped and the retained count is reported with the estimates. The tables report the bootstrap standard error, the 95 percent percentile interval, and $\Pr(> 0)$, the bootstrap probability that the estimate is positive. The probability is one-sided because every hypothesis is directional.

⁶Standardized education, numeracy, and age; indicators for white, female, Democrat, and liberal; the automation-risk percentile; the squares of education, numeracy, and age; and the interactions of numeracy with education, age with white, age with education, and white with education. The Democrat and liberal indicators mark the lower three categories of the seven-point party-identification and ideology scales.

⁷The weighted average in arm d is $\hat{\mu}_d = \sum_{i:D_i=d} p_i \eta_i / \sum_{i:D_i=d} p_i$, where p_i is the fitted probability and η_i the policy-demand score. Normalizing by the realized sum of the weights, rather than by the sample size, makes each arm’s estimate a proper weighted mean, invariant to the scale of the weights; this normalized form is the Hájek estimator.

A second specification adjusts the weighted contrast for ideology, trait anxiety, and age. The adjustment matters because the weighting changes the composition of each arm. In a finite sample the two weighted groups can differ on particular characteristics even though assignment was random, and part of any gap in policy demand would then reflect those characteristics rather than the treatment. The balance checks show differences in these three covariates. The adjustment removes from policy demand the part that the three covariates explain, removes the same from treatment status, and re-estimates the contrast on what remains.⁸ Both removals use the same weights as the contrast itself. The bootstrap refits the adjustment in every replicate. The tables report both versions.

3.6 High-Activation Subgroup Specification

H3 and H4 concern the subgroup in which trait anxiety and automation risk are both high. The test is a pooled Hájek-weighted interaction OLS on the full sample,

$$\eta_i = \alpha + \tau_{\text{base}} D_i + \gamma C_i + \delta (D_i \times C_i) + \varepsilon_i, \quad (2)$$

where C_i indicates the high-activation subgroup: it equals one when both trait anxiety and automation risk, measured pre-treatment, fall in the top tercile of the pooled sample. The model is fit with weights p_i for the high-trust stratum and $1 - p_i$ for the low-trust stratum, the same weights as the stratum contrasts above. τ_{base} is the treatment effect outside the subgroup, δ is the additional effect inside it, and the subgroup effect is $\tau_{\text{base}} + \delta$. No cutoff or sharpness parameter is estimated; the only estimated quantities are the treatment effects. Standard errors follow the same bootstrap as above.

4 Results

4.1 Sample Characteristics and Balance

Of the 901 respondents who completed the experiment, 475 were assigned to treatment and 426 to control. Table 1 reports covariate means and standardized differences by arm for the thirteen pre-treatment variables. No covariate differs between arms after Holm correction. Trait anxiety reaches nominal significance (SMD = 0.14, uncorrected $p = 0.031$) but does not survive correction (adjusted $p = 0.400$). With thirteen simultaneous tests at $\alpha = .05$, one false positive is expected by chance. The observed imbalance is substantively small and is addressed by covariate adjustment in the estimation.

⁸Formally, with weights $w_i = p_i$, policy demand η_i and the treatment indicator D_i are each regressed by weighted least squares on a constant, ideology, trait anxiety, and age. The adjusted estimate is the slope from the weighted regression of the η -residuals on the D -residuals. By the Frisch–Waugh–Lovell theorem this slope equals the coefficient on D_i in the weighted regression of η_i on D_i and the three covariates, and with the covariates omitted it collapses exactly to the Hájek contrast $\hat{\mu}_1 - \hat{\mu}_0$. The adjusted and unadjusted estimators therefore share the same weights, the same target stratum, and the same identification argument, and differ only by the imbalance correction. An ordinary regression with controls would weight respondents differently and would not target the same stratum contrast.

Table 1: Did randomization balance the two arms?

Variable	Treated mean	Treated SD	Control mean	Control SD	Δ	SMD	p
Age	40.080	10.758	38.812	10.439	1.268	0.120	0.073
Ideology	3.173	1.757	3.371	1.848	-0.198	-0.110	0.100
Education	3.726	0.910	3.634	0.906	0.093	0.102	0.127
Numeracy score	2.497	0.713	2.460	0.776	0.037	0.049	0.461
Party ID (7pt)	3.078	2.077	3.197	2.152	-0.119	-0.056	0.399
Locus of control	3.432	0.708	3.443	0.696	-0.011	-0.016	0.806
Trait anxiety	3.763	1.607	3.532	1.586	0.230	0.144	0.031
Economic worry (std.)	0.035	1.024	-0.044	0.966	0.079	0.079	0.235
Expert automation risk (0–100)	42.328	30.825	40.941	30.725	1.387	0.045	0.500
Female	0.516	0.500	0.512	0.500	0.004	0.008	0.903
White	0.701	0.458	0.681	0.467	0.020	0.044	0.511
Risk tolerance index	-0.053	0.986	-0.010	1.012	-0.042	-0.042	0.538
Patience index	-0.000	0.990	0.008	1.027	-0.008	-0.008	0.911

Arm means and standard deviations. Δ = treated – control; SMD = standardized mean difference (Cohen’s d , pooled SD); p from Welch two-sample t -tests. Trait anxiety is nominally imbalanced ($p = 0.031$, SMD = 0.14) but does not survive Holm correction across the 13 covariates (corrected $p = 0.40$).

Three checks confirm the treatment operated as intended (treated arm only, $n = 475$). Comprehension of the expert-assessment method was high: 93.5% of treated respondents answered the factual check correctly.⁹ The perceived gap between own belief and expert estimate tracks the objective Belief Gap closely (Spearman $\rho = -0.74$, $p < 0.001$).¹⁰ Self-reported belief updating also tracks the gap (mean = 3.15 on a five-point scale, SD = 1.22; Spearman $\rho = 0.66$ with the objective gap, $p < 0.001$).¹¹ Respondents understood the information, perceived the discrepancy the design created, and reported updating in proportion to it.

4.2 Unconditional Average Treatment Effect (H1)

Table 2 reports the average treatment effect on the standardized policy-demand composite across five covariate-adjustment specifications. The unadjusted estimate is 0.284 SD (95% CI [0.153, 0.414], $p < 0.001$). Adding ideology, trait anxiety, and age as controls reduces it to 0.230 SD [0.114, 0.346], consistent with the trait-anxiety imbalance noted above absorbing a small share of the raw difference. The effect is positive and statistically clear under every specification. Figure 1 shows that no single item drives the composite: the largest movements are on unemployment benefits (0.25 SD) and labor protections (0.21 SD); the smallest is on retraining (0.10 SD, $p = 0.10$). H1 predicted a modest aggregate effect. The observed effect is modest in absolute terms and statistically significant, but H1 also predicted that the aggregate understates the responding subgroup. That prediction is tested next.

⁹Q136: “According to the information provided, what kind of task is easiest for AI to automate?” Four response options; correct answer: “Tasks that are routine, clearly defined, and follow repeated patterns.”

¹⁰Q141: “How does the experts’ automation risk assessment compare to your initial view?” Response options run from “significantly higher relative risk” (1) to “significantly lower relative risk” (5). The correlation is negative because a positive objective gap (expert estimate above own belief) maps to the low end of this scale.

¹¹Q143: “Compared to your original view, do you now feel your job is at higher or lower risk of automation?” (1 = Much lower than I originally thought; 5 = Much higher than I originally thought).

Table 2: Does personalized risk information raise policy demand on average?

Scale	Controls	ATE	SE	<i>t</i>	95% CI	<i>p</i>
std	No controls	0.284	0.067	4.259	[0.153, 0.414]	0.000
	+ Ideology	0.233	0.060	3.915	[0.117, 0.350]	0.000
	+ TraitAnx	0.258	0.065	3.943	[0.130, 0.387]	0.000
	+ TraitAnx + Ideology	0.223	0.059	3.775	[0.107, 0.339]	0.000
	+ TraitAnx + Ideology	0.230	0.059	3.880	[0.114, 0.346]	0.000
	+ Age					
raw	No controls	0.326	0.077	4.259	[0.176, 0.476]	0.000
	+ Ideology	0.268	0.068	3.915	[0.134, 0.402]	0.000
	+ TraitAnx	0.297	0.075	3.943	[0.149, 0.444]	0.000
	+ TraitAnx + Ideology	0.256	0.068	3.775	[0.123, 0.390]	0.000
	+ TraitAnx + Ideology	0.264	0.068	3.880	[0.131, 0.398]	0.000
	+ Age					

Average treatment effect on policy demand, by covariate-adjustment specification. Additive OLS with HC1 robust SE; ATE = Treated coefficient. Controls (trait anxiety, ideology, age) are mean-centered. $N = 901$. std = standardized composite; raw = mean of the five 1–7 policy items.

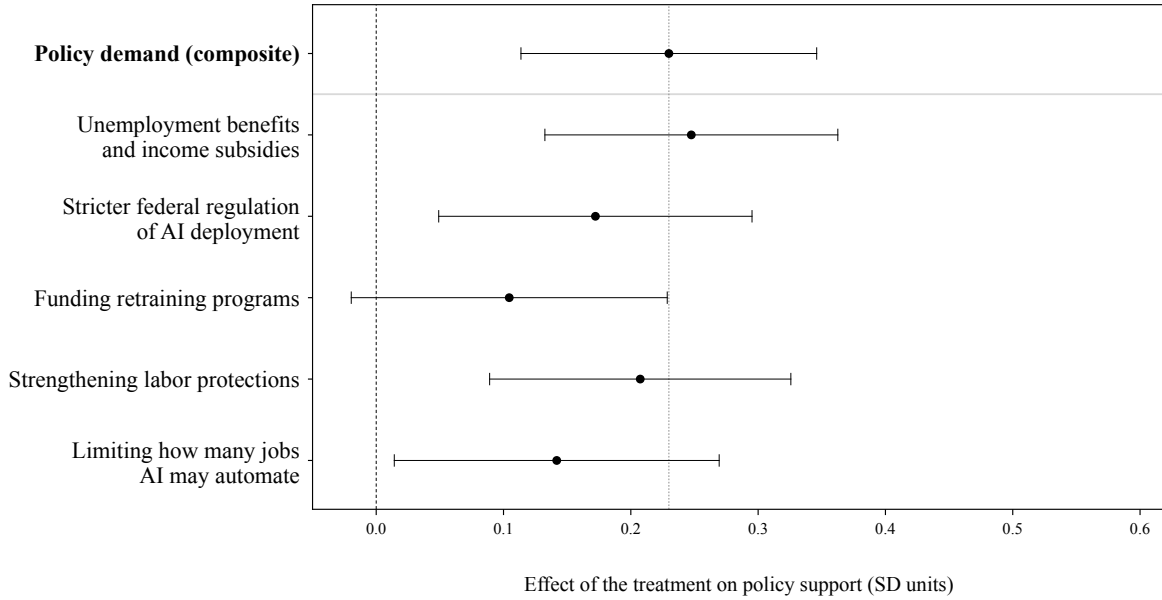


Figure 1: ATE on the standardized policy-demand composite and on each policy item (item outcomes standardized over the full sample). Adjusted specification (trait anxiety, ideology, age; HC1 robust); 95% confidence intervals. The dotted rule marks the composite ATE.

4.3 Trust-Stratum Treatment Effects (H2–H4)

Table S1 reports treatment effects on policy demand by trust stratum, estimated by the principal-score weighting described in Section 3.5. The effect among respondents predicted to trust the source is 0.402 SD (95% CI [0.221, 0.608], $\Pr(> 0) = 1.000$). The effect among the remaining respondents is 0.137 SD [−0.016, 0.269]. The difference between strata is 0.265 SD [0.052, 0.506], $\Pr(> 0) = 0.993$. Adjusting for ideology, trait anxiety, and age reduces the estimates modestly (Table S2): the high-trust CATE falls to 0.329 SD [0.186, 0.543], the low-trust CATE to 0.110 SD [−0.030, 0.226], and the contrast to 0.219 SD [0.057, 0.466]. The reading is the same under both specifications, which Figure 2 displays side by side. Respondents predicted to trust the source respond roughly three times as strongly as those who do not, and the unweighted ATE sits between the two stratum effects, consistent with the composition prediction of H1.

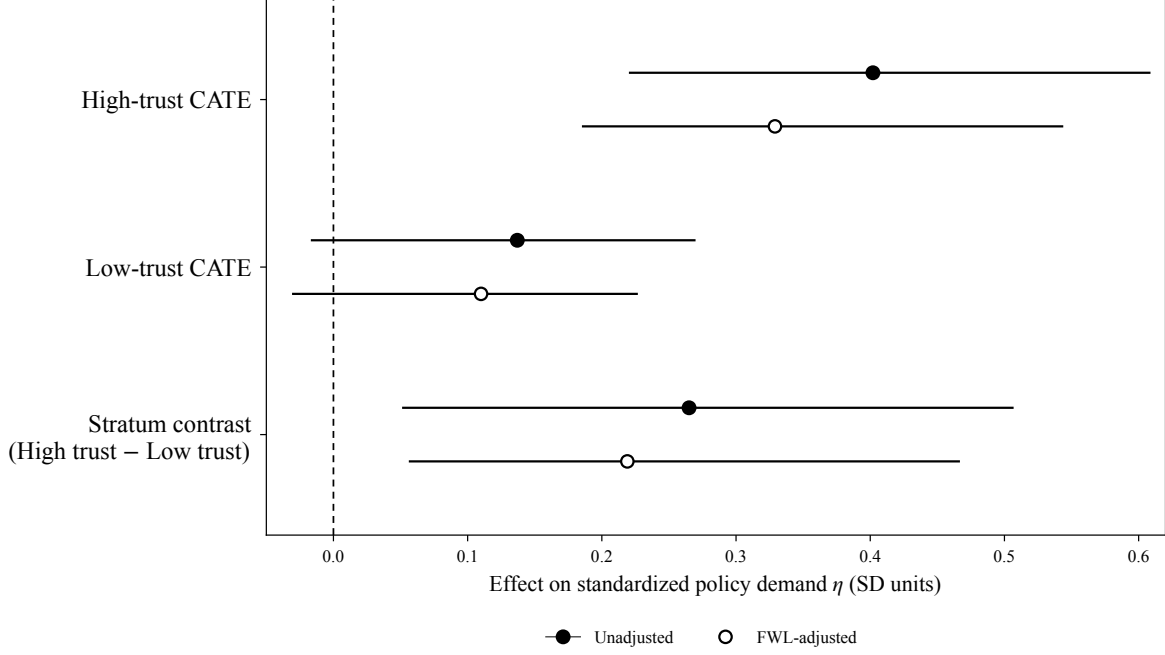


Figure 2: Trust-stratum treatment effects on standardized policy demand (η) under both specifications, for each estimand of Tables S1 and S2, with 95% bootstrap percentile intervals. Both specifications use the same Hájek principal-score weights; the unadjusted estimate (filled) weights the raw outcomes, and the FWL-adjusted estimate (open) additionally partials ideology, trait anxiety, and age out of outcome and treatment.

H3 and H4 predict that the trust-stratum effect is largest where trait anxiety and automation risk are both high. Table 3 reports the interaction model of Section 3.6: the treatment effect outside the high-activation subgroup, τ_{base} , and the additional effect inside it, δ , within each trust stratum. The subgroup contains 105 of 901 respondents (11.7%). Among high-trust respondents, $\tau_{base} = 0.288$ SD [0.128, 0.442] and $\delta = 0.397$ SD [0.083, 0.690], $p = 0.006$; the subgroup effect is $\tau_{base} + \delta = 0.685$ SD [0.364, 0.980]. Among the low-trust stratum, $\delta = 0.194$ SD [-0.056, 0.466], indistinguishable from zero. The result replicates at two alternative cutoffs (median and 58.3rd percentile). The additional effect among high-trust respondents excludes zero at every cutoff tested and decreases toward the stratum average as the subgroup widens. The low-trust additional effect is indistinguishable from zero throughout. The joint effect is present only among respondents who clear the trust gate.

Table 3: Is the treatment effect concentrated in a high-anxiety, high-risk corner? Fixed conjunctive cutoffs at the median, the 58.3rd percentile, and the tercile.

Cutoff	Stratum	τ_{base} (95% CI)	δ (95% CI)	$\tau_{base} + \delta$ (95% CI)	Corner share	$P(\delta < 0)$
Median	High trust	0.275 [0.111, 0.431]	0.265 [0.028, 0.481]	0.540 [0.303, 0.762]	25.1%	0.014
	Low trust	0.102 [-0.022, 0.225]	0.046 [-0.122, 0.214]	0.148 [-0.022, 0.310]	25.1%	0.317
58.3rd pct	High trust	0.285 [0.125, 0.442]	0.288 [0.041, 0.523]	0.573 [0.321, 0.813]	19.6%	0.013
	Low trust	0.091 [-0.034, 0.211]	0.103 [-0.079, 0.281]	0.194 [0.007, 0.379]	19.6%	0.135
Tercile (primary)	High trust	0.288 [0.128, 0.442]	0.397 [0.083, 0.690]	0.685 [0.364, 0.980]	11.7%	0.006
	Low trust	0.092 [-0.033, 0.217]	0.194 [-0.056, 0.466]	0.285 [0.041, 0.552]	11.7%	0.057

A respondent is in the corner if trait anxiety and automation risk, both pre-treatment, each clear the same percentile cutoff in the pooled sample. Cutoffs: the median (50th percentile), the 58.3rd percentile (the midpoint between the median and the top tercile), and the top tercile (66.7th percentile; primary, 105/901 respondents, 11.7%, unweighted). τ_{base} is the effect outside the corner; δ is the additional effect inside it; $\tau_{base} + \delta$ is the total effect for respondents in the corner, bootstrapped as its own quantity. The implied stratum ATE, $\tau_{base} + \delta \cdot \text{share}$, is nearly constant across cutoffs (high trust 0.334–0.336; low trust 0.113–0.114) and reproduces the stratum estimates. Estimates from weighted least squares at each cutoff, with no estimated cutoff or sharpness parameter; SE and CI from a full nonparametric bootstrap ($B = 2000$ per cutoff per stratum, reweighting refit each replicate).

Three features of Table 3 support this reading. The cutoffs are fixed in advance: the median gives a coarse top-half

corner, the top tercile gives the primary definition, and the 58.3rd percentile sits midway between the two, added as an arbitrary intermediate point. The two corner variables are nearly independent, with $\text{corr}(A, R) = 0.014$. Independent terciles would place about 11.1% of the sample in the corner; the observed share is 11.7%. The corner is therefore a genuine joint region, not a product of two correlated screens. The implied stratum ATE, $\tau_{\text{base}} + \delta \cdot \text{share}$, checks the decomposition against the stratum estimates. It is nearly constant across cutoffs, 0.334 to 0.336 for high trust and 0.113 to 0.114 for low trust, and it sits close to the adjusted stratum effects of Table S2 (0.329 and 0.110). The decomposition redistributes each stratum effect between the corner and its complement; it does not manufacture effect.

4.4 Sensitivity and Power

The trust-stratum estimates assume principal ignorability. Latent stratum membership does not predict baseline policy demand beyond the predicted trust score. Table 4 reports the violation magnitude at which each estimate equals zero. The sensitivity parameter Δ is the unobserved difference in treatment response between strata at the same predicted trust score, in standard-deviation units of policy demand. The high-trust CATE equals zero at $\Delta^* = -1.167 [-2.585, -0.837]$. The stratum contrast equals zero at $\Delta^* = -0.515 [-1.421, -0.135]$. Both breakdown values are negative. Overturning either estimate requires latent low-trust respondents to respond more strongly than latent high-trust respondents at the same predicted trust score, the reverse of every observed moderation pattern. The high-trust CATE is robust to large violations. The stratum contrast is robust to moderate violations.

Table 4: How large a violation of principal ignorability would erase the results?

Target	Estimate	Breakdown Δ^*	SE	95% CI
High-trust CATE	0.402	-1.167	0.441	[-2.585, -0.837]
Stratum contrast (High trust - Low trust)	0.265	-0.515	0.328	[-1.421, -0.135]

The sensitivity parameter Δ is the within-proxy-cell difference in treatment effects between latent strata (SD units of η); PI implies $\Delta = 0$, and level violations cancel exactly because both arms share the same weights. Exact bias of the plain Hájek estimator: $\text{bias}(\hat{\tau}_T) = -\Delta E[p(1-p)]/E[p]$ and $\text{bias}(\hat{C}) = -\Delta\lambda$ with $\lambda = E[p(1-p)]/(E[p]E[1-p]) = 0.514$. The breakdown value Δ^* is the violation that drives the target to zero; a violation of this sign means latent low-trust respondents out-respond latent High-trust respondents at identical proxy values, the reverse of every observed moderation pattern. SE and CI from the full bootstrap ($B = 2000$; RF proxy, principal-score logit, weights, and multipliers refit each replicate). Does not cover the FWL-adjusted variant, whose multiplier differs.

Table 5 reports each headline estimate’s minimum detectable effect (MDE) at 80% power. The high-trust CATE exceeds its MDE under both specifications (observed 0.402 vs. 0.277 unadjusted, 0.329 vs. 0.249 adjusted), with power at least 96%. The stratum contrast falls below its MDE (observed 0.265, MDE 0.328 unadjusted, power 62%). The contrast is statistically significant in this sample, but the design is underpowered for an effect of that size. A difference of two estimates has larger sampling variance than either estimate alone. The design is adequately powered for the high-trust CATE and for the high-activation subgroup effect (0.685 SD, power 98%).

Table 5: Which of the trust-stratum estimates are adequately powered?

Target	Observed	Committed SE	MDE	Power at observed
High-trust CATE (unadjusted)	0.402	0.099	0.277	98.1%
High-trust CATE (FWL-adjusted)	0.329	0.089	0.249	95.9%
Stratum contrast (unadjusted)	0.265	0.117	0.328	61.9%
Stratum contrast (FWL-adjusted)	0.219	0.106	0.297	54.2%

MDE: minimum detectable effect at 80% power, $\alpha = 0.05$ (two-sided), computed from the committed SE. SEs are the full-bootstrap values reported in Tables S1 and S2; each replicate refits the RF proxy, the principal-score model, and the Hájek weights. Power at observed: power if the true effect equals the observed estimate. Figure S3 reports the full power curves and a simulation cross-check for the High-trust CATE.

4.5 Effects on Perceived Economic Impacts

CAM’s mechanism runs through perceived exposure to harm: the signal raises policy demand by raising perceived risk. The two intermediate-outcome batteries described in Section 3.3 let this first link be checked directly. The treatment raises job insecurity by 0.14 SD [0.01, 0.27] and perceived negative impacts of AI by 0.23 SD [0.10, 0.35] (adjusted specification); perceived positive impacts stay at zero (-0.02 [$-0.14, 0.11$]). At the item level, six of the eight threat-side items move, between 0.15 and 0.23 SD, and none of the three benefit items does (SI Section S2, Figures S1 and S2). The asymmetry matches the mechanism: the expert signal shifts beliefs about harm to one’s own position, not a general reweighting of AI’s prospects. These estimates describe treatment effects on post-treatment beliefs, in their own right. They are not a mediation analysis: decomposing the policy-demand effect into a share running through these beliefs would require the beliefs to be as good as randomly assigned given covariates, and this design does not support that assumption.

5 Discussion

Our theoretical framework predicts that demand for government protection against automation risk rises only where respondents trust the source and are disposed to act on what it reports. We find an unconditional treatment effect that is modest, 0.23 SD, consistent with H1’s prediction that pooling responsive and unresponsive respondents dilutes the aggregate. The effect among respondents predicted to trust the source is 0.40 SD; among the remaining stratum it is 0.14 SD; the difference between the two groups is 0.27 SD and is identified under randomization and principal ignorability. Within the trust stratum, the effect concentrates further: among respondents whose trait anxiety and perceived automation risk both fall in the top tercile, the additional effect on top of the trust-only baseline brings the total to 0.69 SD, while the corresponding increment among the remaining stratum is indistinguishable from zero. The four predictions hold in the order the model states them, from the aggregate to the joint condition.

The conditional pattern in our results matches the shape of the successful findings in the experimental literature. Where prior automation-information experiments find increased support for government protection, the increase is conditional on characteristics of the receiver. Heinrich and Witko (2025) find that information about AI’s labor-market impact raises support for AI regulation among respondents with high prior technology knowledge and lowers it among those with low knowledge, a 0.34 SD gap around a null average. Jeffrey (2021) finds that fairness rhetoric added to the risk message raises support for corporation tax, top income tax, and unemployment benefit where the risk message alone does not, an effect concentrated among respondents with a high-school, technical-college, or bachelor’s education rather than the least- or most-educated. Gallego et al. (2022) find that prompting workers to reflect on how technology will affect their own future job raises support for slowing technology adoption only among those with high predicted concern, and on no other policy channel. Arntz, Blesse and Doerrenberg (2024) document opposing responses conditional on prior beliefs about technology that cancel in the aggregate.

The manipulation matters as much as the characteristics of the receiver. In Druckman’s Generalizing Persuasion framework, source, message content, and receiver jointly determine whether a message is scrutinized or handled

through a simple cue, and scrutiny takes effort (Druckman, 2022). The occupation-keyed designs differ in the effort their sequences invite. Zhang (2022) pipes the occupation’s probability into a short passage; beliefs up-date, policy preferences do not. Golin and Rauh (2023) elicit the respondent’s prior, show the piped probability, and state the difference between the two; preferences for redistribution rise with that difference, driven by respondents who receive worse news than they expected. Our design explains how experts assess automation risk, elicits the prior, and then sets the expert estimate against it; policy demand rises on average and concentrates where the model predicts. These designs differ on more than one dimension, so no comparison isolates a single feature. Read together, they are consistent with the account CAM makes explicit. Information about automation risk shifts policy demand where the receiver accepts the message and is disposed to act on it, and averaging over heterogeneous responses conceals this structure in the aggregate.

Expert information about automation risk is threatening. Respondents therefore resist what contradicts their prior beliefs about their own prospects (Zaller, 1992). Source credibility governs whether they accept it instead. Citizens substitute a speaker’s account for information they lack when they perceive that speaker as knowledgeable and as sharing their interests (Lupia and McCubbins, 1998), and credibility conditions how framed information is processed (Druckman, 2001). Such trust is politically patterned and stable (Gauchat, 2012; Motta, 2018), so it can be measured before treatment. Together these traditions predict H2. A threatening estimate meets resistance by default, and trust in its source permits acceptance. Two further literatures meet at the second gate. Exposure to economic risk raises demand for government protection (Iversen and Soskice, 2001; Rehm, 2009; Margalit, 2019). Trait anxiety disposes people toward vigilance, under which threatening information receives deliberative processing (Marcus, Neuman and MacKuen, 2000; Valentino et al., 2008). Together they predict H3 and H4. Demand rises with the risk communicated, and most steeply among the anxious. Albertson and Gadarian (2015) find that anxious citizens increase their reliance on trusted experts. That finding places the second gate behind the first. H1 follows from the other three, since an unweighted average dilutes an effect conditional in this way.

The main qualification concerns the trust measure. Respondents can only rate the expert source after seeing its estimate, so the rating exists in the treated arm alone and arrives after treatment. The estimator was built for this. It forms its weights from pre-treatment characteristics only, and the stratum effects are identified under randomization and principal ignorability. The assumption itself is untestable in this design, so we measured how far it can fail instead. A violation large enough to overturn either estimate would have to reverse every moderation pattern the data show. The subgroup result carries a similar safeguard. Its rule was fixed before estimation, and the effect appears at every cutoff we tried. The one estimate this sample cannot settle is the stratum contrast, which is significant here but measured at 62% power.

Our claims are bounded by the sample and the design. The evidence comes from U.S. workers recruited online, in one risk domain, from one credentialed source, with policy preferences stated minutes after the information arrived. Trust measured before treatment gives the control arm its own ratings and collapses the sensitivity analysis to a point. A larger sample settles the stratum contrast. Golin and Rauh (2023) varied the stated source and found no average difference; a design that measures each respondent’s trust in the source it faces would test the first gate directly. Kuo, Manzano and Gallego (2024) find that job loss attributed to robots raises support for compensating the displaced as much as job loss attributed to offshoring, and that only offshoring raises support for preventing the change. The demand we measure warrants the same test in other risk domains.

6 Conclusion

Political economy scholarship predicts increased support for state protection among workers who face new economic risks. Yet, automation-risk experiments have often found little movement in policy demand, and we argued the two findings are compatible. Building on leading theories of information processing, we posited a theoretical synthesis in which two conditions decide the response. Expert information must first be accepted, and acceptance runs through trust in its source. It must then be worth acting on, which depends on the size of the risk and on the worker’s own disposition toward it. Our experiment delivered a credentialed expert estimate of each worker’s own automation risk. Demand for government protection concentrated where both conditions

held.

The mechanism we describe is not specific to automation. Climate projections, pandemic guidance, and economic forecasts are all expert claims about risks people cannot verify for themselves. Our account suggests that such warnings move policy preferences most where recipients trust the experts and are willing to act on what they report. It also implies that average treatment effects understate the response of persuadable respondents. Furthermore, to the extent that trust in experts aligns with people's own politics, then the influence of an expert warning about material, prospective risks depends as much on the standing of its source as on its content.

Thus, a model of conditional activation of expert risk information implies that automation's political consequences depend on which citizens trust the experts issuing the warnings. If expert trust on this domain is divided along political lines, then risk information will activate demand for protection in one part of the electorate and leave the rest unmoved. Hence, the aggregate response will look weaker than the exposure that produced it. Expert credibility is therefore a second axis of the politics of technological change, alongside the scale of displacement.

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Supplemental Information

The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

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S1 Identification of the Trust-Stratum Effects

This appendix states the assumptions under which the weighting estimator of Section 3.5 identifies the trust-stratum treatment effects, proves the identification result, derives the breakdown value reported by the sensitivity analysis, and situates the construction relative to principal-score methods for observed post-treatment variables.

S1.1 Setup

Notation follows Section 3.4. Respondents $i = 1, \dots, N$ ($N = 901$) are randomly assigned $D_i \in \{0, 1\}$ (475 treated, 426 control). Policy demand has potential values $\eta_i(1)$ and $\eta_i(0)$, with $\eta_i = \eta_i(D_i)$ observed. The trust rating $Q_i \in \{1, \dots, 5\}$ is the rating a respondent gives the expert estimate when treated; it is realized only in the treated arm. The stratum indicator $S_i = \mathbf{1}\{Q_i \geq 4\}$ is defined for every respondent through the rating they would give if treated. Because S_i is a function of a potential response, it is unaffected by assignment and functions as a pre-treatment attribute in the sense of principal stratification (Frangakis and Rubin, 2002); only its realization is confined to the treated arm.

Let $\mathbf{Z}_i \in \mathbb{R}^{15}$ collect the fifteen pre-treatment terms listed in Section 3.5. The trust proxy is the fitted value of a random forest (500 trees, maximum depth 10, minimum leaf size 5) trained on the treated arm to predict Q_i from $\mathbf{Z}_i$, evaluated at every respondent's covariates and standardized over the full sample; call it $\hat{Q}_i$. The principal score is the one-dimensional logistic index

$$p_i = \Pr(S_i = 1 \mid \hat{Q}_i) = \text{logit}^{-1}(\gamma_0 + \gamma_1 \hat{Q}_i), \quad (3)$$

with (γ_0, γ_1) estimated by maximum likelihood in the treated arm, where S_i is observed, and the fitted probability computed for every respondent. Because every input to $\hat{Q}_i$ is pre-treatment, both $\hat{Q}_i$ and p_i are defined in both arms.

S1.2 Assumptions

Assumption 1 (Randomization). $D_i \perp\!\!\!\perp (\eta_i(0), \eta_i(1), S_i, \mathbf{Z}_i)$.

Randomization balances $\mathbf{Z}_i$ across arms, so the proxy $\hat{Q}_i$ has the same conditional meaning in the control arm as in the treated arm even though its training target Q_i is realized only in the treated arm.

Assumption 2 (Relevance). $\gamma_1 \neq 0$ in (3): the proxy predicts stratum membership. This is testable in the treated arm.

Assumption 3 (Principal ignorability). For $d \in \{0, 1\}$,

$$\mathbb{E}[\eta_i(d) \mid S_i, \hat{Q}_i] = \mathbb{E}[\eta_i(d) \mid \hat{Q}_i]. \quad (4)$$

At a given value of the proxy, stratum membership carries no further information about the mean of either potential outcome. The $d = 0$ statement carries the cross-arm load: it is what permits the control arm, where S_i is never realized, to inform $\mathbb{E}[\eta(0) \mid S = 1]$. The $d = 1$ statement is also required, because the treated arm enters the estimator weighted by p_i rather than split on the realized indicator; it licenses reading the weighted treated mean as the stratum mean. The sensitivity analysis below parameterizes violations of exactly this pair of restrictions.

Assumption 4 (Overlap). There exist constants $0 < \underline{p} \leq \bar{p} < 1$ with $\underline{p} \leq p_i \leq \bar{p}$ for all i .

One caveat attaches to credibility rather than identification. The principal score is fit where S_i is observed and applied to controls, where it is not. Trust in the expert message cannot be elicited without exposing the respondent to the message, so this cross-arm extension is design-based and untestable in this experiment. Under the stated assumptions the estimands are identified; the sensitivity analysis below disciplines the assumption's credibility.

S1.3 Identification

Proposition 1. Define the Hájek means $\mu_d = \sum_{i:D_i=d} p_i \eta_i / \sum_{i:D_i=d} p_i$ for $d \in \{0, 1\}$. Under Assumptions 1–4, $\mu_1 \xrightarrow{p} \mathbb{E}[\eta(1) \mid S = 1]$ and $\mu_0 \xrightarrow{p} \mathbb{E}[\eta(0) \mid S = 1]$, so $\mu_1 - \mu_0$ identifies τ_{high} of equation (1). Replacing p_i with $1 - p_i$ identifies τ_{low} .

Proof. Fix d . By Assumption 1, $\mathbb{E}[p_i \eta_i \mid D_i = d] = \mathbb{E}[p_i \eta_i(d)]$ and $\mathbb{E}[p_i \mid D_i = d] = \mathbb{E}[p_i] = \Pr(S_i = 1)$. Since p_i is a function of $\hat{Q}_i$, iterated expectations and Assumption 3 give

$$\mathbb{E}[p_i \eta_i(d)] = \mathbb{E}\left[p_i \mathbb{E}[\eta_i(d) \mid \hat{Q}_i]\right] = \mathbb{E}\left[\mathbb{E}[\mathbf{1}\{S_i = 1\} \mid \hat{Q}_i] \mathbb{E}[\eta_i(d) \mid \hat{Q}_i]\right] = \mathbb{E}[\mathbf{1}\{S_i = 1\} \eta_i(d)],$$

where the last equality uses (4) to reattach S_i inside the conditional expectation. Dividing by $\mathbb{E}[p_i] = \Pr(S_i = 1)$ yields $\mathbb{E}[\eta(d) \mid S = 1]$, and the Hájek ratio converges to this limit by the law of large numbers and Slutsky’s theorem given Assumption 4. The low-trust result replaces $S_i = 1$ with $S_i = 0$ and p_i with $1 - p_i$ throughout. $\square$

S1.4 Sensitivity: derivation of the breakdown value

Define the within-cell stratum contrast of each potential outcome,

$$\delta_d(q) = \mathbb{E}[\eta(d) \mid S = 1, \hat{Q} = q] - \mathbb{E}[\eta(d) \mid S = 0, \hat{Q} = q], \quad d \in \{0, 1\}. \quad (5)$$

Principal ignorability is exactly $\delta_0 = \delta_1 = 0$. Decomposing $\mathbb{E}[\eta(d) \mid \hat{Q}]$ over strata and repeating the argument of Proposition 1 without imposing (4) gives, for each arm,

$$\mu_d - \mathbb{E}[\eta(d) \mid S = 1] = - \frac{\mathbb{E}[p(1 - p) \delta_d(\hat{Q})]}{\mathbb{E}[p]}.$$

A violation common to both arms therefore cancels exactly, because both arms carry the same weights: the bias of the estimated stratum effect depends only on the differential violation $\Delta(q) = \delta_1(q) - \delta_0(q)$, the hidden difference in treatment response between latent strata at the same proxy value, in standard-deviation units of η . For constant Δ ,

$$\text{bias}(\hat{\tau}_{high}) = -\Delta \frac{\mathbb{E}[p(1 - p)]}{\mathbb{E}[p]}, \quad \text{bias}(\hat{\tau}_{high} - \hat{\tau}_{low}) = -\Delta \lambda, \quad \lambda = \frac{\mathbb{E}[p(1 - p)]}{\mathbb{E}[p] \mathbb{E}[1 - p]}. \quad (6)$$

The breakdown value Δ^* is the violation at which the true target is zero given the observed estimate:

$$\Delta_{high}^* = -\hat{\tau}_{high} \frac{\mathbb{E}[p]}{\mathbb{E}[p(1 - p)]}, \quad \Delta_{contrast}^* = -\hat{C}/\lambda, \quad (7)$$

with $\hat{C}$ the estimated stratum contrast. The formulas are closed-form and are evaluated at the sample moments of the fitted score, $\mathbb{E}[p] = 0.330$, $\mathbb{E}[p(1 - p)] = 0.114$, and $\lambda = 0.514$; confidence intervals come from the same full bootstrap as the estimates ($B = 2000$, with the random forest, the logit, the weights, and the multipliers refit in every replicate). The algebra is exact for the plain Hájek estimator and does not cover the FWL-adjusted variant, whose bias multiplier differs. Section 4 reports the resulting values (Table 4): both breakdown values are negative, so overturning either estimate requires latent low-trust respondents to respond more strongly than latent high-trust respondents at identical proxy values, the reverse of every observed moderation pattern; the theory’s own prediction is $\Delta > 0$, under which the reported estimates understate the contrast.

S1.5 Relation to principal-score methods

The identifying assumption belongs to the principal-ignorability family of Ding and Lu (2017), and the estimator shares their device of weighting by the principal score divided by the marginal stratum mass. The design premises differ. In their setting the intermediate variable is observed in whichever arm realizes it, and monotonicity assumptions make an observed cell of the treated arm an unmixed sample from a single stratum; their weighting

estimator accordingly leaves the treated arm unweighted and reweights only the control arm. Here the stratum indicator is realized in the treated arm only, no monotonicity restriction is imposed, and both arms are weighted symmetrically by the same score built from pre-treatment covariates. The score also conditions on the scalar proxy $\hat{Q}_i$ rather than on the full covariate vector: the forest compresses the covariates into the single dimension along which trust varies predictably, and principal ignorability is stated given that index. Ding and Lu (2017) is therefore the source of the assumption family and the weighting device, not of the estimator as implemented.

S2 Supplementary Treatment-Effect Estimates

The exhibits in this section supplement the estimates of Section 4. Tables S1 and S2 report the trust-stratum estimates that Figure 2 displays: the Hájek principal-score estimates and their FWL-adjusted variant, with bootstrap standard errors, 95% percentile intervals, and $\Pr(> 0)$ for each estimand.

Table S1: Does the treatment effect run through trust in the source?

Estimand	Est.	SE	95% CI	P(>0)
High-trust CATE	0.402	0.099	[0.221, 0.608]	1.000
Low-trust CATE	0.137	0.073	[-0.016, 0.269]	0.960
Stratum contrast (High trust - Low trust)	0.265	0.117	[0.052, 0.506]	0.993

Trust-stratum effects on standardized policy demand (η), Hájek principal-score weighting (Ding & Lu 2017), identified under randomization and principal ignorability. Full bootstrap $B = 2000$ (RF proxy, principal-score logit, and Hajek weights refit each replicate; η frozen at EAP scores). Unweighted ATE on $\eta = +0.229$. $N = 475$ treated, 426 control.

Table S2: Do the trust-stratum effects survive adjustment for residual imbalance?

Estimand	Hájek	FWL adj.	Shift	SE	95% CI	P(>0)
High-trust CATE	0.402	0.329	-0.072	0.089	[0.186, 0.543]	1.000
Low-trust CATE	0.137	0.110	-0.027	0.065	[-0.030, 0.226]	0.936
Stratum contrast (High trust - Low trust)	0.265	0.219	-0.046	0.106	[0.057, 0.466]	0.995

Randomization balances covariates in the raw sample, but the principal score p_i is a function of covariates, so weighting by it re-tilts the arms and can leave or amplify chance imbalance (Table S4). The FWL estimator partials $Z = [\text{ideology, trait anxiety, age}]$ out of both η and T by weighted least squares on $[1, Z]$, then regresses residual on residual: $\hat{\tau} = \sum_i w_i \tilde{T}_i \tilde{\eta}_i / \sum_i w_i \tilde{T}_i^2$. This equals the coefficient on T in the weighted regression of η on $[1, T, Z]$, and collapses to the plain Hájek contrast when Z is dropped, so the two columns differ only by the imbalance correction. SE and CI from the same full bootstrap as Table S1 ($B = 2000$; RF proxy, principal-score logit, weights, and the residualization all refit each replicate). $N = 475$ treated, 426 control.

Table S3 disaggregates the composite policy-demand ATE to the individual item level, across the same covariate specifications, so that the composite result can be checked against the items driving it.

Table S3: Which policy items drive the composite effect?

Controls	Unemp. benefits	AI regulation	Retraining	Labor protections	Limit automation
No controls	0.495*** (0.109)	0.308** (0.094)	0.213* (0.090)	0.349*** (0.098)	0.264* (0.107)
+ Ideology	0.409*** (0.096)	0.256** (0.089)	0.165 (0.085)	0.282** (0.089)	0.228* (0.105)
+ TraitAnx	0.467*** (0.109)	0.273** (0.092)	0.198* (0.090)	0.321** (0.097)	0.224* (0.105)
+ TraitAnx + Ideology	0.405*** (0.096)	0.239** (0.088)	0.164 (0.085)	0.273** (0.089)	0.202 (0.104)
+ TraitAnx + Ideology + Age	0.406*** (0.096)	0.243** (0.089)	0.141 (0.085)	0.304** (0.088)	0.227* (0.104)

Per-item ATEs on the raw 1–7 item scales. Each cell reports ATE (HC1 SE) with significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Controls (trait anxiety, ideology, age) are mean-centered and match the composite-table specs.

Table S4 reports the diagnostic behind the adjustment set of the FWL specification. Randomization balances covariates between arms in expectation, but the weighting can undo part of that balance in a finite sample. The weight p_i is a function of pre-treatment covariates, so weighting each arm by it re-tilts the arms along those covariates. The table quantifies the tilt. Each entry is the treated-minus-control difference in the weighted mean of a standardized covariate, in SD units. The unweighted differences are modest: -0.109 for ideology, 0.144 for trait anxiety, 0.120 for age. Weighting toward the high-trust stratum amplifies two of them: the ideology difference grows to -0.157 and the trait-anxiety difference to 0.212 , while the age difference falls to 0.077 . Weighting toward the low-trust stratum moves the three less: -0.075 , 0.106 , and 0.149 . These three covariates form the adjustment set. The FWL adjustment removes their contribution from the weighted contrast, and Table S2 reports the result.

Table S4: How imbalanced is the weighted sample, and on which covariates?

Covariate	Unweighted	High trust weighted	Low trust weighted
Ideology	-0.109	-0.157	-0.075
Trait anxiety	0.144	0.212	0.106
Age	0.120	0.077	0.149

Treated minus control difference in the (weighted) covariate mean; all three covariates are standardized, so entries are in SD units. Weighting toward High-trust respondents (p_i) amplifies the ideology and trait-anxiety imbalance relative to the unweighted sample, which is what the FWL adjustment in Table S2 removes.

Table S5 tests whether the trust result depends on partisanship and ideology inside the proxy. The canonical forest includes a Democrat indicator and a liberal indicator among its fifteen terms. The worry is that the proxy identifies partisans rather than trusters. The check drops the two indicators and changes nothing else; no interaction involves either term, so exactly two of the fifteen terms leave. The reduced pipeline is refit end to end, with the same full bootstrap. The reduced weights track the canonical ones closely (Pearson correlation 0.845). The estimates barely move. The high-trust CATE is 0.424 against the canonical 0.402 , and the stratum contrast is 0.295 against 0.265 ; under the FWL adjustment the shifts are smaller still. Intervals that excluded zero under the canonical proxy still exclude it here. Party and ideology are not the source of the trust-stratum result.

Table S5: Does the trust result depend on partisanship and ideology in the proxy?

Weighting	Estimand	Reduced RF	Canonical	Shift	SE	95% CI	P(>0)
Hájek	High-trust CATE	0.424	0.402	0.022	0.100	[0.272, 0.659]	1.000
	Low-trust CATE	0.129	0.137	-0.008	0.074	[-0.031, 0.257]	0.937
	Stratum contrast (High trust - Low trust)	0.295	0.265	0.030	0.118	[0.106, 0.572]	0.998
FWL adj.	High-trust CATE	0.326	0.329	-0.003	0.089	[0.180, 0.534]	1.000
	Low-trust CATE	0.114	0.110	0.003	0.064	[-0.029, 0.229]	0.941
	Stratum contrast (High trust - Low trust)	0.213	0.219	-0.006	0.105	[0.040, 0.456]	0.992

The reduced random forest drops the Democrat and liberal indicators from the canonical 15-term feature set (no interaction involves either, so exactly two terms are removed) and the principal-score pipeline is otherwise unchanged. The resulting weights track the canonical ones closely: $\text{corr}(p_i^{\text{red}}, p_i) = 0.845$ (Pearson; Spearman 0.858). SE and CI from the full bootstrap of the reduced pipeline ($B = 2000$; reduced RF, principal-score logit, weights, and the FWL residualization refit each replicate); the canonical column repeats the point estimates of Tables S1 and S2, whose uncertainty is reported there.

Figures S1 and S2 report treatment effects on the two intermediate-outcome batteries of Section 4’s design: the four-item Job Insecurity and Automation Beliefs scale (Q94) and the seven-item Perceived Impacts of AI scale (Q95), whose items load on a negative-impacts factor (F1) and a positive-impacts factor (F2). At the item level, the treatment moves the threat-side items: concern about losing one’s job (0.17 SD), ten-year automation likelihood (0.15), reduced job opportunities (0.20), lower salary (0.18), decreased job security (0.23), and worsened work conditions (0.20). The three positive-impact items do not move. The scale level shows the same separation under both specifications: job insecurity rises 0.14 SD (adjusted), perceived negative impacts rise 0.23, and perceived positive impacts stay at zero ($-0.02, [-0.14, 0.11]$). The information shifts perceived exposure to harm; it does not shift beliefs about AI’s benefits.

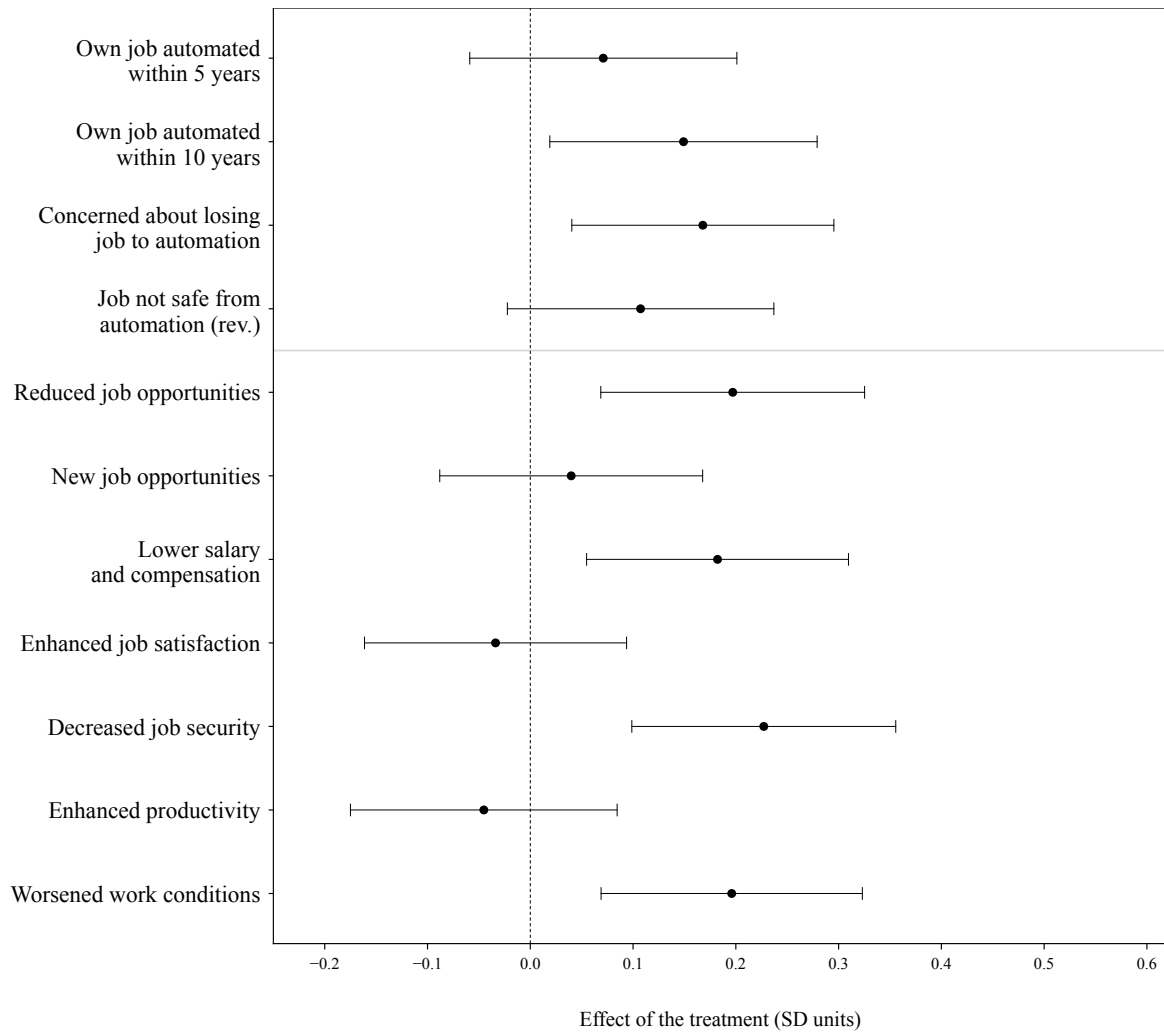


Figure S1: Treatment effects on each intermediate-outcome item, standardized over the full sample: the Job Insecurity and Automation Beliefs items (Q94, above the separator) and the Perceived Impacts of AI items (Q95, below). Adjusted specification (trait anxiety, ideology, age; HC1 robust); 95% confidence intervals.

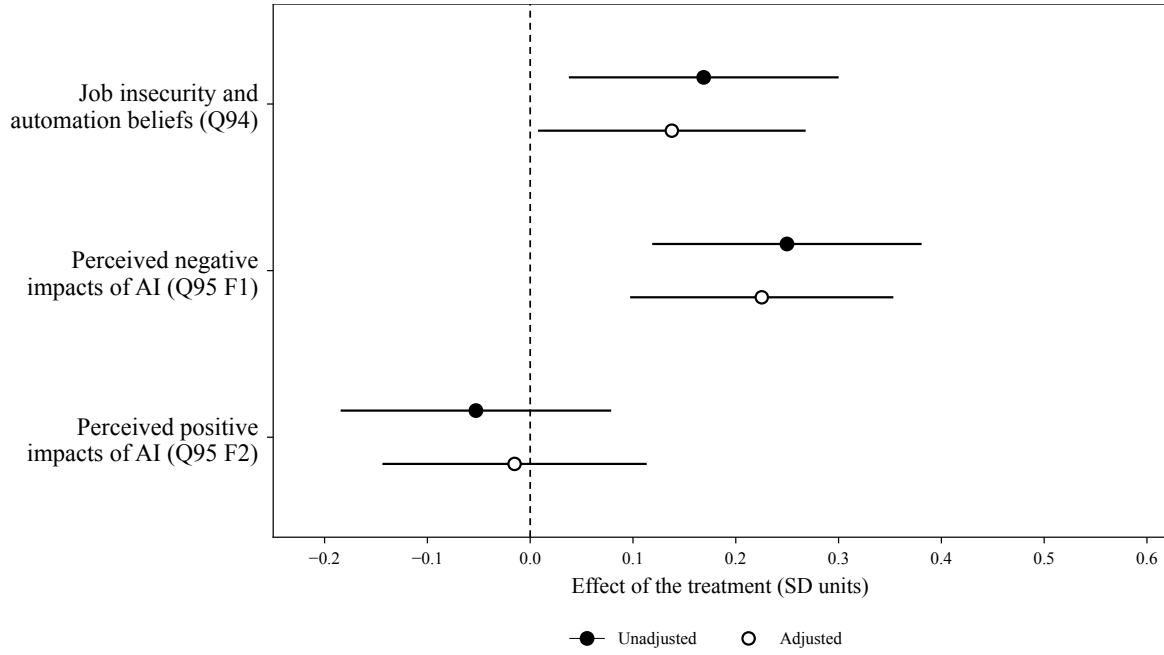


Figure S2: Treatment effects on the three intermediate-outcome scales, standardized: the Q94 scale mean and the two Q95 factor means (negative impacts F1: items 1/3/5/7; positive impacts F2: items 2/4/6). Filled: unadjusted OLS; open: adjusted for trait anxiety, ideology, and age (HC1 robust); 95% confidence intervals.

S3 Power and Minimum Detectable Effects

S3.1 Trust-Stratum Estimates

Table 5 in the main text reports the minimum detectable effect (MDE, 80% power) for the high-trust CATE and the stratum contrast, each under both specifications, using the SE the full bootstrap reports (Tables S1 and S2), not a re-derived or simplified one. Figure S3 reports the full power curves. The high-trust CATE is comfortably powered under both specifications: the observed effect sits well above its own MDE. The stratum contrast is not: its observed effect falls below its own MDE under both specifications, a predictable consequence of comparing two noisy estimates rather than reading one directly.

Figure S3 shows a simulated cross-check for the high-trust CATE only. The asymmetry is deliberate. The simulation injects a candidate true effect into resamples of the real data, refits, and records how often the effect is detected. This design needs a target that equals a single coefficient in a single weighted regression. The high-trust CATE meets that condition: weighted least squares of policy demand on treatment, weighted by the principal score, returns exactly the Hájek contrast. The stratum contrast does not. It is the difference of two estimates that weight the same outcomes differently (p_i against $1 - p_i$), so no single injected effect produces a chosen true contrast; the injection would have to be rescaled by the empirical distribution of the weights. That extra machinery would add fragility to a diagnostic. The analytic curve already covers the contrast, and it rests on the committed bootstrap SE. The simulation also holds the principal score fixed, which understates that SE. Its role is to check the curve's shape for the better-powered target, not to replace the committed uncertainty for either target.

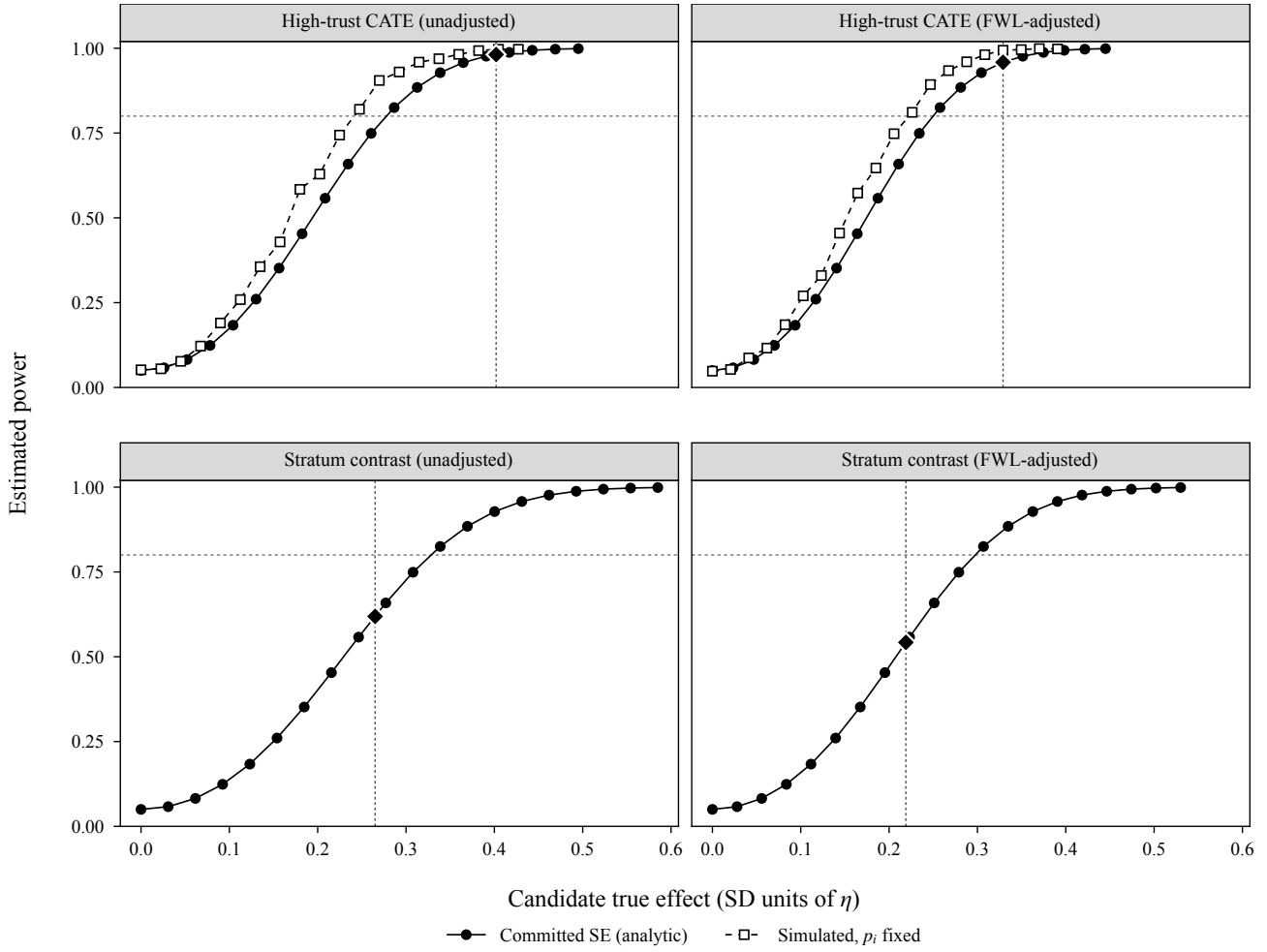


Figure S3: Power curves for the high-trust CATE and the stratum contrast, both specifications. The high-trust CATE additionally shows a simulated cross-check (case-resampling with effect injection, principal score held fixed); that simulation understates the committed bootstrap’s SE by 12–14% since it does not refit the RF proxy and logit, so its curve is, if anything, optimistic relative to the committed one.

S3.2 The Conjunctive Corner

A null or marginal δ is ambiguous between “no corner effect” and “an effect too small for this design to reliably detect” unless power is addressed directly. Table S6 reports the minimum detectable effect for four targets at 80% power: each stratum’s δ (the corner increment), and each stratum’s corner effect ($\tau_{base} + \delta$, the total effect for respondents actually in the corner). The MDE is the smallest true effect this design would catch 80% of the time on repeated sampling. All four targets come from the same pooled, Hájek-weighted interaction model reported in the main text, with the corner targets obtained by reparameterizing its treatment terms so that $\tau_{base} + \delta$ is a single coefficient rather than a linear combination; the fit is unchanged, and the corner point estimates therefore reproduce Table 3 exactly. The MDE is computed two ways, and they agree closely: a textbook normal-approximation formula, and a simulated power curve (Figure S4) that resamples the real data and injects a candidate true effect on top of the actual covariate distribution and residual structure, rather than assuming an idealized one. The corner effects are the better-powered quantities, since they are larger than the increments δ against essentially the same noise. The high-trust corner effect is the one target clearing its own 80% MDE, at 98% estimated power; the low-trust corner effect reaches 59%. The increments themselves both fall short (72% for high trust, 32% for low trust), so a null reading on the low-trust δ in particular cannot distinguish a genuinely absent corner effect from a real one this design lacks the power to see reliably.

Table S6: How small a true effect could this design have reliably found?

Target	n	Observed	MDE (analytic)	MDE (bootstrap)	Power at observed
delta: High trust	901	0.397	0.426	0.445	72.4%
delta: Low trust	901	0.194	0.364	0.360	32.4%
Corner ($\tau_{base} + \delta$): High trust	901	0.685	0.440	0.461	98.0%
Corner ($\tau_{base} + \delta$): Low trust	901	0.285	0.368	0.360	58.8%

The minimum detectable effect (MDE) is the smallest true effect this design would detect with 80% probability at $\alpha = 0.05$ (two-sided). “Analytic” is the textbook $(z_{1-\alpha/2} + z_{power}) \times SE$ using the target’s own robust (HC1-style) analytic SE; “bootstrap” comes from a genuine simulated power curve: case-resample the real data, inject a candidate true effect on top of the OTHER regressors’ fitted contribution plus the real residual (so the real covariate distribution and heteroskedasticity are preserved, not an idealized textbook DGP), refit, and record the detection rate across 1000 replicates per grid point (Figure S4). “Power at observed” is the estimated power AT the estimator’s own observed effect size: not whether the result is significant in this sample, but how reliably a design like this one would detect an effect of that exact size on repeated sampling. All four targets come from the same pooled interaction model on the full sample with the stratum’s Hájek weight, the model reported in Table 3; the Corner rows (the total effect $\tau_{base} + \delta$ for respondents actually in the corner) simply reparameterize the treatment terms $\{T, T \times \text{Corner}\}$ as $\{T \times (1 - \text{Corner}), T \times \text{Corner}\}$, which spans the identical column space and leaves the fit numerically unchanged but makes $\tau_{base} + \delta$ a single coefficient, so their point estimates reproduce the Corner column of Table 3 exactly. The two Corner effects are much better powered than the increments δ : High trust’s Corner effect reaches 98% power and is the one target clearing its own 80% MDE, because it is a larger effect relative to essentially the same noise as δ alone. The remaining three fall below their MDE, most sharply Low trust’s δ (32% power): a null reading there cannot distinguish “no corner effect” from “a real effect this design lacks the power to see reliably.”

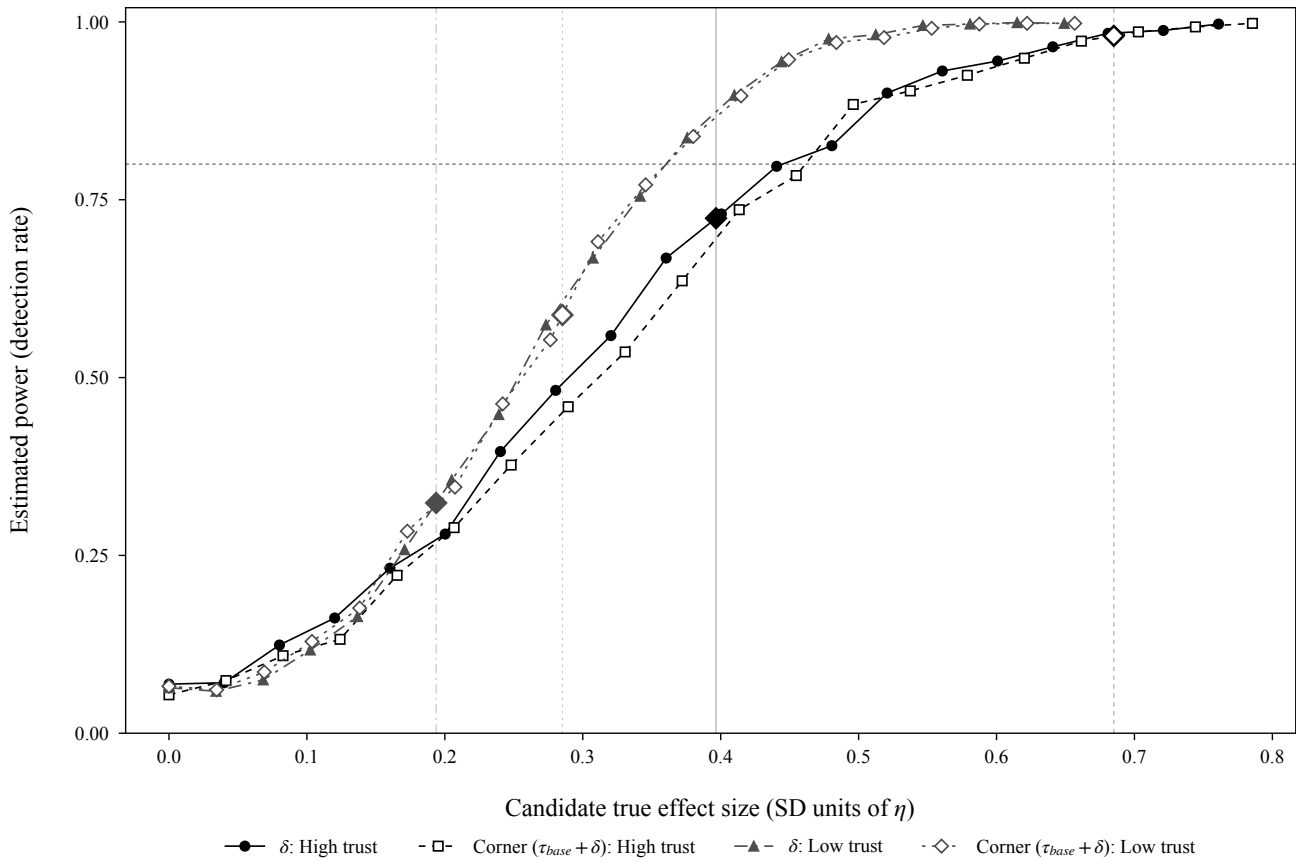


Figure S4: Bootstrapped power curves for δ and the total corner effect, both strata (series identified in the legend). Diamonds mark each target’s own observed effect size and its estimated power there; the dotted horizontal line is the 80% power target.

S4 Definition of the Conjunctive Corner

Corner membership is a deterministic function of two pre-treatment variables, so the rule can be drawn directly on the data it partitions. The primary tercile cutoff and the two coarser cutoffs used in the sensitivity columns of Table 3 are shown together.

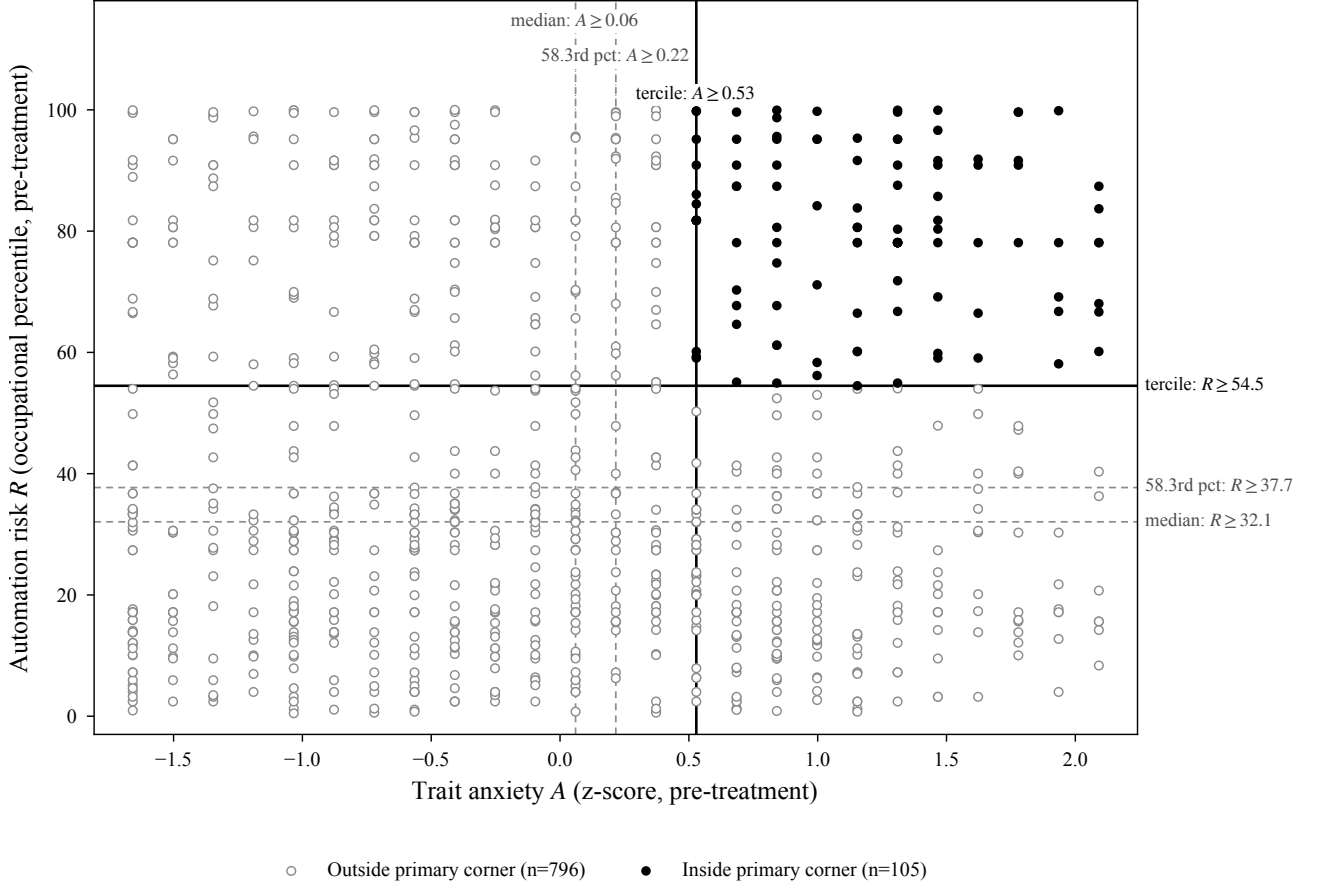


Figure S5: The corner as defined: trait anxiety vs. automation risk for all 901 respondents, with the primary tercile cutoff (solid) and the two sensitivity cutoffs (dashed) drawn on both axes.

S5 Robustness: Flexible Threshold Estimation

As a robustness check on the fixed-tercile test of Section 4, this appendix lets the corner's cutoffs and sharpness be estimated from the data rather than fixed a priori. The model is a two-class latent mixture in which the classes differ only in treatment sensitivity. The outcome model is $y_i = \alpha + Z_i\beta + (\tau_{base} + \delta S_i)T_i + \varepsilon_i$, where S_i indicates the sensitive class and membership is a conjunctive threshold in trait anxiety A and automation risk R : $\Pr(S_i = 1) = \text{logit}^{-1}\{\kappa(A_i - a^*)\} \cdot \text{logit}^{-1}\{\kappa(R_i - r^*)\}$, with both cutoffs estimated rather than imposed (Table S7).

The model is estimated by Hamiltonian Monte Carlo with the No-U-Turn sampler in Stan (Carpenter et al., 2017), fit separately to each trust stratum: four chains of 1,500 warmup and 1,500 sampling iterations each, 6,000 posterior draws per fit. Point estimates are posterior medians, intervals are central 95 percent credible intervals from the posterior quantiles, and $\Pr(\delta < 0)$ is the posterior mass below zero. Every fit reported in this appendix ran without divergent transitions, with a maximum split $\hat{R}$ of 1.008.

Three constraints carry the interpretation. The sensitivity increment δ is floored at -0.1 , just below zero, so the posterior can bank against the boundary when the classes coincide and a positive δ is evidence rather than an artifact of a non-negativity constraint. The sharpness κ is floored where the logistic transition would span

the entire observed support, since below that the specification degenerates into a linear ramp rather than a corner; a fixed- κ sensitivity grid (gradual, distinct, near-hard step) is reported in Table S8 because the data cannot pin down sharpness and separation simultaneously at this sample size. The principal score enters as a pseudo-likelihood weight (p_i for the high-trust stratum, $1 - p_i$ for the low-trust stratum), rescaled to the Kish effective sample size so the posterior reflects the information the design carries. The reconstruction $\tau_{base} + \delta \cdot \text{share}$ must reproduce the stratum ATE estimated by the other methods; that identity is the model’s internal sanity check.

The percentile-scale variant re-expresses A and R as in-sample percentile ranks in $[0, 1]$, the scale on which the thresholds have a structural domain: a cutoff in the sample distribution lies in $[0, 1]$ by construction. This affords strictly truncated priors, $a^*, r^* \sim \mathcal{N}(0.5, 0.25)$ on $[0, 1]$, centered at the median with 95% prior mass approximately $[0.06, 0.94]$. No observations are trimmed. The sharpness floor and sensitivity grid are rescaled to the unit support (transition widths of one half, one fifth, and one tenth of the support), reported in Table S9, so the variant differs from the benchmark only in scale and prior, not in structure; it does not repeat the separability check of SI Section S6, which covers the benchmark specification only.

Table S7: Does the corner hold when the cutoffs are estimated rather than fixed?

Fit	τ base	δ	δ 95% CrI	$P(\delta < 0)$	Corner	a^* pct	r^* pct	Share	Implied ATE
High trust	0.290	0.409	[0.006, 0.959]	0.023	0.698	0.699	0.636	0.105	0.334
Low trust	0.081	0.359	[-0.030, 1.053]	0.042	0.430	0.835	0.647	0.072	0.113
High trust, sharpness fixed = 2	0.288	0.378	[-0.006, 0.900]	0.028	0.664	0.648	0.647	0.123	0.336
High trust, sharpness fixed = 5	0.292	0.419	[0.033, 0.995]	0.015	0.708	0.740	0.633	0.097	0.334
High trust, sharpness fixed = 10	0.294	0.435	[0.021, 1.025]	0.018	0.724	0.835	0.626	0.091	0.333

Two latent classes differing only in treatment sensitivity, membership a conjunctive threshold $\Pr(\text{sensitive}) = \text{logit}^{-1}(\kappa(A - a^*)) \cdot \text{logit}^{-1}(\kappa(R - r^*))$ with the cutoffs *estimated*, not imposed. τ base is the effect outside the corner, δ the additional effect inside it, Corner their sum. A conjunctive form is used because a logistic linear index has a planar decision boundary and cannot represent the hypothesis: fitted that way, the posterior for δ reproduced its prior. δ is floored at -0.1 rather than at 0, so its posterior can bank against the boundary when the classes coincide and a positive estimate is evidence rather than an artifact of a non-negativity constraint. Rows below the rule hold corner sharpness κ fixed, since the data cannot pin down sharpness and separation simultaneously. **Sanity check:** For the High trust stratum: $\text{ATE}_{tr} = \tau_{btr} + \delta_{tr} \mathbb{E}[\pi_i]$. For the Low trust stratum: $\text{ATE}_{ml} = \tau_{bml} + \delta_{ml} \mathbb{E}[1 - \pi_i]$. These implied ATEs must reproduce the stratum ATE obtained by every other estimator here (≈ 0.33 High trust, ≈ 0.11 low-trust); they do. **Calibration:** across 60 simulated homogeneous datasets run through the identical procedure, the 95% CrI excluded zero in 1 (rate 0.017, 95% CI [0.000, 0.089], against a nominal 0.025), and the null δ exceeded the observed value in 0.017 of them.

Table S8: Where do the estimated cutoffs place the corner, and how large is the effect there?

Quantity	High trust	Low trust
τ_{base} (95% CI)	0.290 [0.131, 0.455]	0.081 [-0.054, 0.206]
δ (95% CrI)	0.409 [0.006, 0.959]	0.359 [-0.030, 1.053]
Corner (95% CrI)	0.698 [0.314, 1.264]	0.430 [0.067, 1.139]
a^* percentile	69.9%	83.5%
r^* percentile	63.6%	64.7%
Corner share	10.5%	7.2%

Corner = $\tau_{base} + \delta$. Rows give τ_{base} and δ with 95% CIs, the resulting Corner estimate with its 95% CrI, the estimated percentile location of the anxiety (a^*) and risk (r^*) cutoffs, and the share of respondents falling into the corner, for each stratum. Percentiles are reported as percentages; Corner share is the fraction of the sample in the corner.

Table S9: Does the estimated-cutoff corner hold on the percentile scale?

Quantity	High trust	Low trust
τ_{base} (95% CI)	0.291 [0.130, 0.448]	0.087 [-0.047, 0.213]
δ (95% CrI)	0.356 [0.010, 0.875]	0.244 [-0.060, 0.852]
Corner (95% CrI)	0.642 [0.299, 1.189]	0.323 [0.036, 0.959]
a^* percentile	59.4%	69.9%
r^* percentile	62.3%	60.2%
Corner share	12.8%	10.4%
Sample share (unw.)	13.6%	10.3%

Percentile-scale specification with truncated threshold priors. Anxiety and risk enter as in-sample percentiles in $[0, 1]$; no observations are trimmed. The thresholds a^* and r^* carry $\mathcal{N}(0.5, 0.25)$ priors truncated to $[0, 1]$, their structural domain on the percentile scale. Rows give τ and δ with 95% intervals; Corner = $\tau + \delta$, for each stratum. Sample share (unw.) reports the unweighted posterior-averaged fraction of respondents in the corner.

Agreement with the fixed-tercile test. For the high-trust stratum the two approaches agree closely: τ_{base} 0.288 (fixed tercile) vs. 0.291 (flexible, percentile-scale), δ 0.397 vs. 0.356, implied ATE 0.336 vs. 0.337. The fixed-tercile δ interval is comfortably narrower and clears zero more decisively (95% CI [0.083, 0.690] vs. [0.010, 0.875]), because it pays no estimation cost for the corner’s location; all sampling variability is spent on δ itself. The low-trust stratum is null under both specifications. This agreement across a fixed, assumption-light rule and a flexible, data-estimated one is why the fixed-tercile test is reported as primary.

S6 Robustness: Independence, Identification, and Separability

The conjunctive threshold specification of SI Section S5 does not assume that trait anxiety and automation risk are independent. $\Pr(\text{sensitive} \mid A_i, R_i)$ is evaluated at each respondent’s observed pair, so the joint distribution of A and R never enters the likelihood. The product is a functional form for a response surface, not a factorisation of a joint density into $f(a \mid r)f(r)$. A logistic regression on two correlated covariates makes no independence claim for the same reason.

Dependence threatens identification instead. The conjunction “needs both” is distinguishable from “needs either” only when the off-diagonal quadrants are populated. Here $\text{corr}(A, R) = 0.014$, and the four quadrants at the estimated cutoffs hold 425, 242, 148, and 86 respondents. The distinction is identified.

The assumption open to testing is separability: whether the anxiety cutoff moves with risk, $a^*(R) = a_0 + a_1 R$. The primary model imposes $a_1 = 0$; Table S10 relaxes it. The a_1 row shows that the data carry no information about separability: its posterior reproduces its $N(0, 0.5)$ prior (prior 95% interval $\approx [-0.98, +0.98]$), and only 0.34 of posterior mass falls within ± 0.2 . The informative comparison is between the two δ rows. Freeing a_1 moves δ by 0.010. The estimate survives integration over complete ignorance about separability, a stronger claim than having verified separability itself.

Table S10: Does the conjunctive form require trait anxiety and automation risk to be independent?

Quantity	Estimate	95% CrI
a_1 : anxiety cutoff shift per SD of risk	-0.103	[-0.953, 0.814]
δ , separability imposed ($a_1 = 0$)	0.409	[0.006, 0.959]
δ , separability relaxed	0.399	[-0.003, 0.936]
Implied ATE, separability relaxed	0.337	

The two δ rows carry the comparison: the primary model imposes separability ($a_1 = 0$); the relaxed model lets the anxiety cutoff vary with risk, $a^*(R) = a_0 + a_1 R$. Freeing a_1 moves δ by 0.010. The a_1 row is reported for completeness; its posterior reproduces its $N(0, 0.5)$ prior. Interpretation in the section text.

S7 Design Validity and Measurement

S7.1 Were Prior Beliefs Informative?

The expert estimate delivered genuine news. Figure S6 plots each treated respondent's elicited prior against the expert percentile for their occupation: the two are nearly unrelated ($R^2 = 0.02$), and the fitted line is far from the calibration diagonal. Figure S7 shows that respondents did not know how uninformed they were. The mean absolute belief gap is flat across declared certainty (left panel; Spearman $\rho = 0.005$, $p = 0.91$). Some respondents declared their prior accurate to within one point; 2% of them were right (right panel). The hit rate rises across the panel only because the declared tolerance widens. Prior beliefs do not track expert risk, and self-assessed accuracy does not track actual accuracy. Respondents therefore could not have corrected their own beliefs; the treatment's third step supplies the correction.

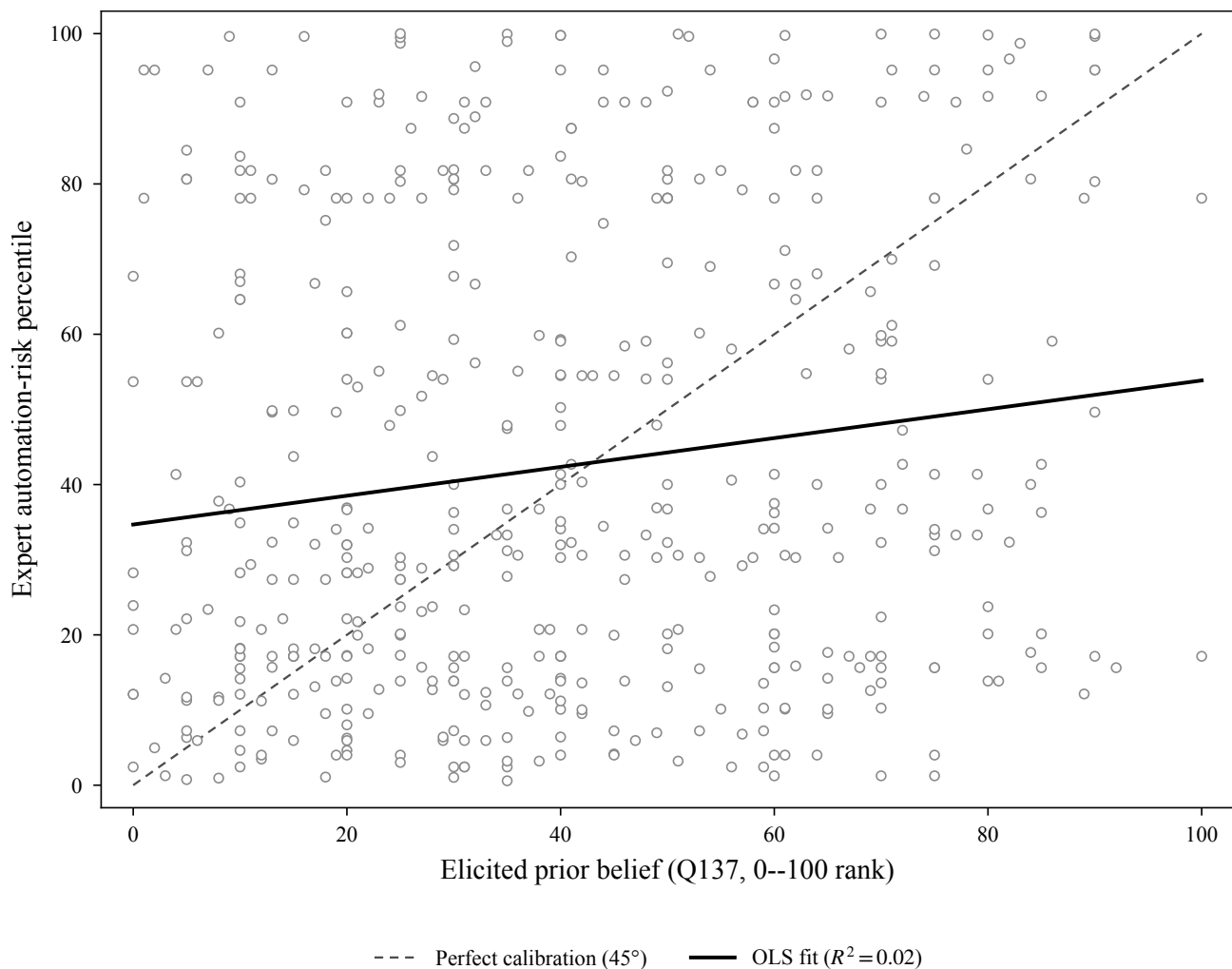


Figure S6: Do prior beliefs track expert automation risk? Treated arm ($n = 475$); the prior is elicited before the expert estimate is revealed. Elicited rank (Q137, 0–100) against the employment-weighted expert percentile; dashed line marks perfect calibration.

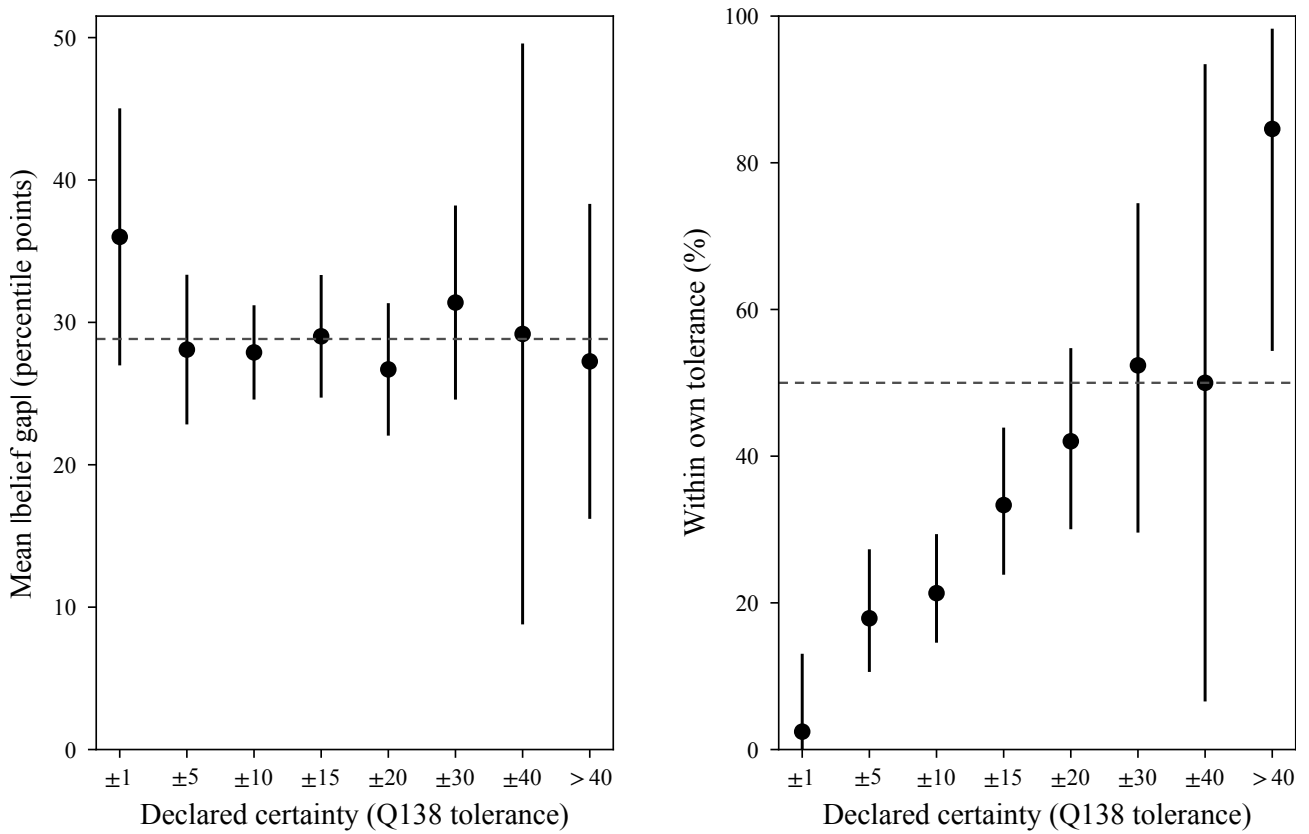


Figure S7: Do respondents know how accurate their priors are? Treated arm ($n = 475$). Left: mean absolute belief gap by declared certainty (Q138 tolerance), with 95% confidence intervals; the dashed rule is the overall mean. Right: share of respondents whose actual absolute gap falls within their own declared tolerance, with Clopper–Pearson 95% intervals; the dashed rule marks 50%.

S7.2 Manipulation Check

Table S11 reports the factual manipulation check among treated respondents. 93.5% answer the comprehension item correctly, and both the perceived gap and the belief update track the objective belief gap (Spearman $|\rho| = 0.74$ and 0.66).

Table S11: Did treated respondents absorb the information?

Check	Statistic	Value	p	n
Comprehension (Q136)	pct_correct	93.500		475
Perceived gap (Q141)	spearman_rho vs Gap	-0.737	0.000	475
Belief update (Q143)	mean	3.145		475
Belief update (Q143)	sd	1.221		475
Belief update (Q143)	spearman_rho vs Gap	0.660	0.000	475

Factual manipulation check, treated arm only. Q136 comprehension pass rate (%); Spearman correlations of perceived gap (Q141) and belief update (Q143) with the objective belief gap.

S7.3 Measurement Validation

Table S12 reports confirmatory factor analysis fit for each multi-item scale. Table S13 reports internal consistency: Cronbach's α ranges from 0.83 (policy demand) to 0.93 (job automation threat). Table S14 reports the per-item loadings behind the fit models.

Table S12: Do the multi-item scales fit as single measurement models?

Scale	<i>N</i>	χ^2	df	<i>p</i>	χ^2/df	CFI	TLI	RMSEA	SRMR
Policy Demand (Q96)	901	31.131	3	0.0000	10.377	0.983	0.944	0.102	0.026
Trait Anxiety (Q23)	901	24.326	1	0.0000	24.326	0.991	0.945	0.161	0.014
Job Automation Threat (Q94)	901	12.114	1	0.0005	12.114	0.997	0.980	0.111	0.006
Automation Impact (Q95, 2-factor)	901	54.672	10	0.0000	5.467	0.987	0.974	0.070	0.026
Locus of Control (Q81)	901	28.855	1	0.0000	28.855	0.903	0.420	0.176	0.041
Economic Worry (Q24/Q25/Q26)	901	0.000	0			1.000			0.000

Confirmatory factor analysis fit indices by scale; semopy ML estimation. Economic Worry is a saturated (df = 0) just-identified model.

Table S13: Are the scales internally consistent?

Scale	<i>k</i>	<i>N</i>	Cronbach's α	McDonald's ω
Policy Demand (Q96)	5	901	0.826	0.832
Trait Anxiety (Q23)	4	901	0.913	0.915
Job Automation Threat (Q94)	4	901	0.930	0.929
Automation Impact F1 (Q95 1/3/5/7)	4	901	0.914	0.914
Automation Impact F2 (Q95 2/4/6)	3	901	0.802	0.805
Locus of Control (Q81)	4	901	0.477	0.522
Economic Worry (Q24/Q25/Q26)	3	901	0.706	0.727

Scale reliability. *k* items; α = Cronbach's alpha; ω = McDonald's omega (from a single-factor fit). Negatively-worded items reverse-scored.

Table S14: How strongly does each item load on its factor?

Scale	Item	Factor	Loading	SE	Std. β	Sig
Policy Demand (Q96)	Q96_1_Policy	F	1.000		0.616	
	Q96_2_Policy	F	0.899	0.057	0.643	***
	Q96_3_Policy	F	0.795	0.046	0.597	***
	Q96_4_Policy	F	1.333	0.078	0.919	***
	Q96_5_Policy	F	1.002	0.064	0.633	***
Trait Anxiety (Q23)	Q23_1_TraitAnx	F	1.000		0.922	
	Q23_2_TraitAnx	F	0.787	0.025	0.831	***
	Q23_3_TraitAnx	F	0.805	0.026	0.796	***
	Q23_4_TraitAnx	F	0.808	0.025	0.833	***
Job Automation Threat (Q94)	Q94_1_AutoAnxiety	F	1.000		0.855	
	Q94_2_AutoAnxiety	F	1.101	0.021	0.854	***
	Q94_3_AutoAnxiety	F	1.163	0.036	0.891	***
	Q94_4_AutoAnxiety_rev	F	1.040	0.034	0.849	***
Automation Impact (Q95, 2-factor)	Q95_1_AutoImpact	F1	1.000		0.789	
	Q95_3_AutoImpact	F1	1.117	0.076	0.918	***
	Q95_5_AutoImpact	F1	1.023	0.031	0.797	***
	Q95_7_AutoImpact	F1	1.085	0.075	0.875	***
	Q95_2_AutoImpact	F2	1.000		0.795	
	Q95_4_AutoImpact	F2	0.922	0.090	0.726	***
	Q95_6_AutoImpact	F2	0.780	0.079	0.638	***
Locus of Control (Q81)	Q81_1_LoC	F	1.000		0.595	
	Q81_2_LoC	F	0.877	0.205	0.380	***
	Q81_3_LoC	F	0.626	0.149	0.378	***
	Q81_4_LoC	F	0.497	0.138	0.260	***
Economic Worry (Q24/Q25/Q26)	Q24_JobWorry	F	1.000		0.594	
	Q25_FinWorry	F	1.615	0.148	0.915	***
	Q26_EconHope	F	0.804	0.063	0.517	***

Confirmatory factor analysis loadings. Loading = unstandardized estimate (first indicator per factor fixed to 1); Std. β = standardized loading. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

S8 Survey Instrument

This appendix reproduces the complete survey instrument, including all question blocks, the experimental treatment stimuli, and outcome measures. Item numbers correspond to the Qualtrics survey flow.

S8.1 Survey Flow Overview

The survey followed this sequence:

1. **Demographics & Screening:** State of residence, occupation selection, attention check.
2. **Political & Personal Background:** Party identification, ideology, trait anxiety, economic anxiety.
3. **Numeracy:** Three-item numeracy battery.

4. **Risk, Time, Locus of Control:** Staircase elicitation tasks (adapted from Falk et al.), locus of control items, prior knowledge of automation.
5. **Experimental Section** (Treatment group only):
 - Information on AI and automation
 - Subjective belief elicitation (automation risk percentile)
 - Expert assessment reveal (Frey & Osborne scores)
 - Factual manipulation checks
6. **Outcome Measures:** Job insecurity beliefs, perceived AI impacts, policy preferences.
7. **Final Demographics:** Gender, race/ethnicity, age, education, religiosity.

S8.2 Demographics and Screening

Q176: In which state do you reside? [Dropdown]

Q177: Please pick the major occupation group that best matches the main kind of work you do. What's an "occupation group"? It's a broad category of related jobs (for example, "Healthcare Support" or "Management"). Select one from the list: [Dropdown: SOC major groups]

Q178: Now, based on the major occupation group you selected, please pick the specific occupation that best matches the main kind of work you do. [Selected major group]: [Conditional dropdown]

Q159: If your specific occupation is not listed, please use the back button to return to the previous question and select a more fitting major occupation group.

Attention Check (Q181–Q182): "So that we know you are paying attention, please ignore the question on screen and select 'Often.'" [Followed by decoy question about dining out]

S8.3 Political and Personal Background

Q19: Generally speaking, do you usually think of yourself as a Democrat, a Republican, an independent, or what?

- Democrat (1)
- Republican (2)
- Independent (3)
- Other party (please specify)

Q20 [If Democrat/Republican]: Would you call yourself a strong [Party] or a not very strong [Party]?

Q21 [If Independent]: Do you think of yourself as closer to the Republican Party or to the Democratic Party?

Q22: When it comes to politics, do you usually think of yourself as...

- Extremely Liberal (1) ...Extremely Conservative (7)

Q23 (Trait Anxiety: Big Five Short Form):

Here are characteristics that may or may not apply to you. Please indicate the extent to which you agree or disagree with each statement. [7-point scale: Strongly disagree to Strongly agree]

1. I see myself as anxious, easily upset.
2. I feel threatened easily.

3. I worry about things.
4. I am afraid of many things.

Q24–Q26 (General Economic Worry):

- Q24: How worried are you about losing your job in the near future? [5-point scale: Extremely / Very / Moderately / A little / Not at all worried]
- Q25: So far as you and your family are concerned, how worried are you about your current financial situation? [5-point scale: Extremely / Very / Moderately / A little / Not at all worried]
- Q26: Thinking about your personal economic well-being, how hopeful do you feel? [5-point scale: Extremely / Very / Moderately / A little / Not at all hopeful]

S8.4 Numeracy Battery

Q29: Imagine that we roll a fair, six-sided die 1,000 times. Out of 1,000 rolls, how many times would the die come up even (2, 4, or 6)? [Open text]

Q30: In the BIG BUCKS LOTTERY, the chance of winning a \$10.00 prize is 1%. If 1,000 people each buy a single ticket, how many people would win? [Open text]

Q31: In the ACME PUBLISHING SWEEPSTAKES, the chance of winning a car is 1 in 1,000. What percentage of tickets win a car? [Open text, 0–100]

S8.5 Risk and Time Preferences

Q150 (Risk Willingness): How willing or unwilling are you to take risks in general? On a scale from 0 to 10, where 0 means you are completely unwilling to take risks and 10 means you are very willing to take risks, how would you rate yourself? Please select a number from 0 to 10: [Dropdown, 0–10]

Q151 (Time Willingness): How willing are you to give up something that is beneficial for you today in order to benefit more from that in the future? On a scale from 0 to 10, where 0 means you are completely unwilling and 10 means you are very willing, how would you rate yourself? Please select a number from 0 to 10: [Dropdown, 0–10]

Staircase Tasks (adapted from Falk et al. for US context):

- **Patience Task:** Five binary choices between receiving \$235 today vs. a larger amount in 12 months. Initial future offer: \$360; bounds: [\$240, \$505]. If participant chooses immediate payment (impatient), future amount increases; if participant chooses future payment (patient), future amount decreases. Iterates 5 times to approximate indifference point.
- **Risk Task:** Five binary choices between a lottery (50% chance of \$700, 50% chance of \$0) vs. a sure payment. Initial sure payment offer: \$375; bounds: [\$25, \$725]. If participant chooses lottery (risk-seeking), sure payment increases; if participant chooses sure payment (risk-averse), sure payment decreases. Iterates 5 times to approximate certainty equivalent.

S8.6 Locus of Control

Q81: Please indicate how much the following statements apply to you. [1–5 scale: Does not apply at all to Fully applies]

1. If I work hard, I will succeed.
2. I'm my own boss.
3. Whether at work or in my private life: What I do is mainly determined by others.

4. Fate often gets in the way of my plans.

S8.7 Prior Knowledge of Automation

Q148: Before today, had you seen, read, or heard anything about how likely your job or occupation is to be automated—that is, replaced in part or whole by computers, AI, or machines? [Yes / No / Not sure]

Q149 [If Yes]: How much had you seen, read, or heard about this topic before today? [A great deal / Some / A little]

Q146 [If Yes]: Where did you get information about how automatable your job or occupation might be? [Select all that apply]

- Employer communication (e.g., newsletter, HR updates)
- Government websites or materials
- News media (TV, newspapers, websites)
- Social media (e.g., Facebook, TikTok, LinkedIn)
- Podcasts
- Internet search you did on your own
- Friends or family
- Somewhere else – please specify
- I don't remember

S8.8 Experimental Treatment (Treatment Group Only)

S8.8.1 Initial Information

Q86: Please carefully read the following information. It will be relevant for the questions ahead.

Automation and Your Job

Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. As AI improves, some jobs may be partly or fully done by machines, while new kinds of jobs also appear.

How do experts judge a job's automation risk?

Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns (and there are lots of examples for computers to learn from), AI systems can pick them up more easily. The shorter and more repetitive a task is, the easier it is to automate.

Example job	Typical task	AI risk (today)
IT Help Desk Technician	10-minute help IT tickets that follow a script	<i>Higher Risk</i> – chatbots can handle most tickets
IT Systems Designer	Plan how the company's tech infrastructure will work over the next year	<i>Lower Risk</i> – needs long-term planning and team buy-in

Other factors also influence how likely a job is to be replaced by AI automation:

Jobs are at greater risk when they:

- Can be done entirely online, or
- Have lots of recorded data for AI to learn from.

Jobs are less at risk when they:

- Require in-person skills or dexterity, or
- Rely on human trust or empathy.

S8.8.2 Factual Manipulation Check

Q136: According to the information provided, what kind of task is easiest for AI to automate?

1. Tasks that require advanced programming skills and high-level technical expertise
2. Tasks that depend on empathy and social connection
3. Tasks that are part of complex computer simulations and advanced research
4. **Tasks that are routine, clearly defined, and follow repeated patterns** [Correct]

S8.8.3 Belief Elicitation

Q137: Imagine every worker in the country standing on one long line, sorted by how likely their job is to be automated. The very front of the line (position 0) belongs to people whose jobs are safer than almost everyone else's, while the very back (position 100) belongs to those whose jobs are at risk sooner than almost everyone else's. Someone standing in the middle of the line (position 50) would have half the number of workers ahead of them and half behind. Safest jobs are those whose chances of being automated are below most other people. Riskier jobs are those with an automation potential above most other people.

Taking a moment to reflect on your own job, where do you feel you would place yourself on that line right now? [Slider: 0–100]

Q138: We now want to get a sense of how certain or uncertain are you in your previous estimate. Thinking about the spot you just placed on the line, about how many positions above or below do you feel it might be off from where experts would put your job—right now?

- **I'm absolutely certain** – No more than 1 spot above or below
- Within 5 spots above or below
- Within 10 spots above or below
- Within 15 spots above or below
- Within 20 spots above or below
- Within 30 spots above or below
- Within 40 spots above or below
- **I'm extremely uncertain** – More than 40 spots above or below

S8.8.4 Expert Assessment Reveal

Q139: Now let's compare your view with what experts think.

Dr. Carl Benedikt Frey and Dr. Michael A. Osborne are researchers at the University of Oxford, where they investigate the impact of AI and automation on the labor market. Their 2017 academic study, "The Future of Employment," is a landmark in this field—cited more than 17,000 times and widely used to gauge automation risk. Here's how Dr. Frey and Dr. Osborne estimated job automation risk: They started with the U.S. government's O*NET database, which lists the skills and day-to-day tasks for every job. A panel of AI researchers first tagged 70 sample jobs as either "automatable" or "not yet automatable." A computer model then learned the patterns: jobs heavy on

routine, rule-based tasks, fine hand-eye work, or data processing score higher; jobs that rely on creative thinking, social insight, or physical dexterity score lower. That model produced a score for each job—the higher the number, the sooner experts think machines could take over most of its tasks. We use that score as the expert estimate for your occupation's relative risk of AI automation.

Your view vs. the expert estimate

Below, you'll see where you placed your job on the automation risk scale, compared to the expert estimate for your occupation.

Q140: [Personalized comparison display rendered via Qualtrics JS.]

Comparing Your Job Automation Risk: Your View vs. Expert Insight

Earlier, we asked you to imagine all workers standing in a single line, ranked by the risk of their jobs being automated, from the lowest relative risk (position 0) to the highest relative risk (position 100).

You placed yourself at: **Position** [SUBJECTIVE SCORE]

Researchers place your job at around: **Position** [OBJECTIVE SCORE]

This means that, according to experts, your job has a **higher** risk of automation than [OBJECTIVE SCORE] out of 100 people, but a **lower** risk than [100 – OBJECTIVE SCORE] others.

S8.8.5 Post-Treatment Checks

Q141 (Manipulation Check): Regarding how likely is your job to be automated in the future relative to others, how does the experts automation risk assessment compare to your initial view?

1. The experts place my job at a significantly higher relative risk
2. The experts place my job at a slightly higher relative risk
3. The experts place my job at a fairly similar relative risk
4. The experts place my job at a slightly lower relative risk
5. The experts place my job at a significantly lower relative risk

Q142 (Trust): To what extent do you think the Oxford researchers are a trustworthy source of information about job automation risk?

1. Not at all trustworthy (1)
2. Slightly trustworthy (2)
3. Moderately trustworthy (3)
4. Very trustworthy (4)
5. Extremely trustworthy (5)

Q143 (Belief Updating): The information you just read provided a risk score for your occupation. Compared to your original view, do you now feel your job is at higher or lower risk of automation?

1. Much lower than I originally thought (1)
2. A little lower than I originally thought (2)
3. About the same as I originally thought (3)
4. A little higher than I originally thought (4)
5. Much higher than I originally thought (5)

S8.9 Outcome Measures

S8.9.1 Job Insecurity and Automation Beliefs

Q94: How much do you agree or disagree with the following statements? [1–7: Strongly disagree to Strongly agree]

1. My job is likely to be automated in the next 5 years.
2. My job is likely to be automated in the next 10 years.
3. I am concerned about losing my job to automation or AI.
4. I feel secure that my occupation is safe from automation.

S8.9.2 Perceived Impacts of AI

Q95: To what extent do you believe that advances in AI will have the following impacts on your career and job prospects? [1–5: Definitely False to Definitely True]

1. Reduced job opportunities: AI will lead to fewer available jobs for me, making it harder to find a job.
2. New jobs: AI-driven innovation will spur economic growth and create new job opportunities for me.
3. Lower salary and compensation: AI will cause a decline in my future wages and benefits in my field.
4. Enhanced job satisfaction: AI's automation of routine tasks will free up my time for creative and strategic work.
5. Decreased job security: AI will make my employment less stable or increase my risk of being laid off.
6. Enhanced productivity: AI will provide tools and assistance that allow me to complete my work more efficiently and effectively.
7. Worsened work conditions: AI will lead to a decline in my pay, benefits, or the quality of my work environment.

S8.9.3 Policy Preferences (Primary Outcome)

Q96: To what extent do you support or oppose the following policy responses to address AI-driven automation's workforce impact? [1–7: Strongly Oppose to Strongly Support]

1. Expanding unemployment benefits and income subsidies.
2. Introducing stricter federal regulations on AI deployment, such as requiring companies to undergo ethical AI reviews.
3. Funding government or private-sector retraining programs to help workers transition to new careers.
4. Strengthening labor protections, such as supporting unionization efforts in AI-exposed industries.
5. Limiting the number of jobs that AI can automate within companies to protect human employment.

S8.10 Final Demographics

Q104 (Gender): Male / Female / Other / Prefer not to say

Q105 (Race/Ethnicity): [Select all that apply] White, Black or African American, Hispanic or Latino, Asian, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, Other, Prefer not to say

Q106 (Age): [Numeric entry]

Q107 (Education): Less than high school / High school diploma / Some college / Bachelor's degree / Graduate degree

Q108 (Religiosity): Thinking about your life these days, do you ever attend religious services, apart from occasional weddings, baptisms or funerals?

1. Every week
2. Almost every week
3. Once or twice a month
4. A few times a year
5. Never, but I am religious
6. Never, I am not religious

S9 A Formal Model of Conditional Activation

This appendix presents a formal counterpart to the framework of Section 2. The goal is not a complete theory of preference formation but a minimal model that clarifies how two predispositions condition the absorption of expert risk information, and that delivers H1 through H4 as signed comparative statics. The architecture follows Canini, González and Prem (2026), who model trust responses to police violence as Bayesian updating that operates only when an activation condition converts a realized event into an effective signal; in their setting the activation condition is media coverage, in ours it is trust in the source. The belief technology follows Little, Schnakenberg and Turner (2022): agents form an optimal conclusion that trades an accuracy motive against a directional motive, in the tradition of Little (2019). Bounded rationality enters only in the belief stage; the action stage is standard expected-utility choice.

S9.1 Setup

A worker i faces latent automation risk $\theta_i \in \mathbb{R}$, an index of displacement probability. The worker holds a prior belief $\theta \sim N(m_i, \sigma_0^2)$. The prior mean m_i summarizes the worker's own informational environment: workers already convinced of high risk have little room to update. Treatment delivers an expert signal $x_i = \theta_i + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma_x^2)$, the credentialed estimate of own-occupation risk. Control receives no signal. Two pre-treatment predispositions complete the setup: trust in expert sources $t_i \geq 0$, and trait anxiety $a_i \geq 0$. Throughout, risk indices are positive ($m > 0, x > 0$), which keeps all action-stage solutions interior.

Standard conjugacy gives the Bayesian benchmark. The posterior after observing x is normal with mean and variance

$$\mu(x) = m + \kappa(x - m), \quad \kappa = \frac{\sigma_0^2}{\sigma_0^2 + \sigma_x^2}, \quad \bar{\sigma}^2 = \frac{\sigma_0^2 \sigma_x^2}{\sigma_0^2 + \sigma_x^2}. \quad (8)$$

S9.2 Belief Stage: Trust-Gated Conclusions

Rather than adopting the posterior mechanically, the worker forms a conclusion $\tilde{\theta}$ about own risk by solving

$$\tilde{\theta}^*(x; t) \in \arg \max_{\tilde{\theta}} \log f_{\theta|x}(\tilde{\theta} | x) + \delta(t) v(\tilde{\theta}), \quad v(\tilde{\theta}) = -(\tilde{\theta} - m)^2. \quad (9)$$

The first term is the accuracy motive of Little, Schnakenberg and Turner (2022): the log density of the Bayesian posterior, which penalizes conclusions that are objectively unlikely. The second term is a confirmation-bias directional motive: a threatening expert signal that contradicts the worker's self-assessment is costly to internalize, so the worker prefers conclusions near the prior mean m . Our departure from Little, Schnakenberg and Turner (2022) is the weight $\delta(t) \geq 0$, which we take to be decreasing in trust, $\delta'(t) < 0$. Resistance to the message falls as credence in the source rises; $\delta(t)$ is a reduced form for the source-credibility channel of Lupia and McCubbins (1998) and Druckman (2001).

Lemma 1 (Posterior conclusion). *Problem (9) has the unique solution*

$$\tilde{\theta}^*(x; t) = m + \omega(t) \kappa(x - m), \quad \omega(t) = \frac{1}{1 + 2\delta(t)\bar{\sigma}^2} \in (0, 1], \quad (10)$$

with (i) $\omega'(t) > 0$; (ii) $\omega \rightarrow 1$ as $\delta \rightarrow 0$; (iii) $\omega \rightarrow 0$ as $\delta \rightarrow \infty$; and (iv) under the threshold form $\delta(t) = \infty \cdot \mathbf{1}\{t < \bar{t}\}$, $\omega(t) = \mathbf{1}\{t \geq \bar{t}\}$.

Proof. By (8), $\log f_{\theta|x}(\tilde{\theta} | x) = -(\tilde{\theta} - \mu)^2/(2\bar{\sigma}^2)$ up to a constant, so the objective is $\Phi(\tilde{\theta}) = -(\tilde{\theta} - \mu)^2/(2\bar{\sigma}^2) - \delta(\tilde{\theta} - m)^2$ with $\Phi'' = -1/\bar{\sigma}^2 - 2\delta < 0$ for all $\delta \geq 0$. Strict concavity makes the first-order condition sufficient and the maximizer unique:

$$-\frac{\tilde{\theta} - \mu}{\bar{\sigma}^2} - 2\delta(\tilde{\theta} - m) = 0 \iff \tilde{\theta}^* = \frac{\mu + 2\delta\bar{\sigma}^2 m}{1 + 2\delta\bar{\sigma}^2}.$$

Substituting $\mu = m + \kappa(x - m)$ yields (10). For (i), $\omega'(t) = 2\bar{\sigma}^2(-\delta'(t))\omega(t)^2 > 0$ because $\delta'(t) < 0$. Claims (ii) and (iii) are immediate from the definition of ω , and (iv) follows by evaluating the two branches. $\square$

Equation (10) carries the first gate in one line. The effective updating weight on the expert signal is the product $\omega(t) \kappa$: an information term κ , the room to update, and a trust term $\omega(t)$, the willingness to accept. At $\delta = 0$ the worker is a full Bayesian, so the rational-expectations benchmark is nested exactly as in Little, Schnakenberg and Turner (2022). For $\delta > 0$ the conclusion shrinks the Bayesian update toward the prior, the pattern Hill (2017) documents experimentally and labels cautious Bayesian updating. Two population-level consequences follow immediately, in the vocabulary of Little, Schnakenberg and Turner (2022): desensitization, because the average updating slope $E[\omega(t)] \kappa$ lies below the Bayesian benchmark κ ; and divergence, because workers who receive the same signal but hold different levels of trust reach different conclusions.

Lemma 2 (Control arm). *A control worker, who receives no signal, concludes $\tilde{\theta}_C^* = m$ for every $\delta \geq 0$.*

Proof. Without a signal the accuracy motive is the log prior density, so the objective is $-(\tilde{\theta} - m)^2/(2\sigma_0^2) - \delta(\tilde{\theta} - m)^2$. Both terms are uniquely maximized at $\tilde{\theta} = m$, hence so is their sum. $\square$

Lemma 2 restates the design's logic inside the model: the confirmation motive is centered at the prior, so with nothing to resist there is no belief distortion in control, and the two arms differ only through the treated arm's gated update.

S9.3 Action Stage: Policy Demand

Given the conclusion $\tilde{\theta}$, the worker chooses policy demand $y \geq 0$, the intensity of support for government protection, to maximize expected utility

$$y^*(\tilde{\theta}; a) \in \arg \max_{y \geq 0} g(a) \tilde{\theta} y - c(y). \quad (11)$$

The benefit of protection scales with the concluded risk $\tilde{\theta}$ and with a vigilance weight $g(a) > 0$, $g'(a) > 0$: trait anxiety raises the decision weight placed on the threat state. The cost $c(y)$ of demanding protection, through taxes, foregone flexibility, or ideological distance, is strictly convex with $c'(0) = 0$. The quadratic case $c(y) = y^2/2$ yields the closed form $y^* = g(a) \tilde{\theta}$; for general convex costs the first-order condition $c'(y^*) = g(a) \tilde{\theta}$ gives $\partial y^*/\partial \tilde{\theta} = g(a)/c''(y^*) > 0$ and $\partial y^*/\partial a = g'(a) \tilde{\theta}/c''(y^*) > 0$, so every sign below survives; the quadratic case removes third-derivative terms from the cross-partials and is the benchmark we report.

S9.4 Comparative Statics and the Hypotheses

Proposition 2 (Individual treatment effect). *Under the quadratic cost, the treatment effect for a worker with signal x , trust t , and anxiety a is*

$$\Delta(x, t, a) = y_T^* - y_C^* = g(a) \omega(t) \kappa (x - m), \quad (12)$$

and therefore

$$(i) \quad \frac{\partial^2 \Delta}{\partial x \partial t} = g(a) \omega'(t) \kappa > 0; \text{ in the threshold case } \partial \Delta / \partial x = 0 \text{ for } t < \bar{t}. \quad (H2)$$

$$(ii) \quad \Delta \text{ is increasing in the communicated discrepancy } (x - m), \text{ with slope } g(a) \omega(t) \kappa. \quad (H3)$$

$$(iii) \quad \frac{\partial^2 \Delta}{\partial x \partial a} = g'(a) \omega(t) \kappa > 0 \text{ and } \frac{\partial^3 \Delta}{\partial x \partial a \partial t} = g'(a) \omega'(t) \kappa > 0. \quad (H4)$$

Proof. Lemma 2 and (11) give $y_C^* = g(a) m$; Lemma 1 and (11) give $y_T^* = g(a) [m + \omega(t) \kappa (x - m)]$. Subtraction yields (12). Claims (i) through (iii) are direct differentiations of this trilinear form, with signs from $g > 0$, $g' > 0$, $\omega > 0$, $\omega' > 0$ (Lemma 1) and $\kappa > 0$; the threshold statement follows from Lemma 1(iv). $\square$

Part (i) is the trust gradient of H2. Part (ii) is the risk-level gradient of H3, and it also locates the workers with little room to move: a high prior m implies a small discrepancy $(x - m)$ and a small effect, the pre-treated margin of Section 2. Part (iii) is the anxiety amplification of H4, and its third cross-partial states something sharper: the amplification is itself increasing in trust, so the largest effect sits where trust, communicated risk, and anxiety coincide, which is the corner tested in Section 4.

Proposition 3 (Aggregation). *Let workers draw predispositions (t_i, a_i) and discrepancies $D_i = x_i - m_i$ with $E[D_i | t_i, a_i] = \Delta \bar{x}$ common across predispositions. Then*

$$\text{ATE} = \kappa \Delta \bar{x} E[g(a) \omega(t)]. \quad (13)$$

Moreover:

(i) *If $\Pr(\delta(t) > 0) > 0$, then $E[\omega(t)] < 1$: the aggregate updating slope lies strictly below the Bayesian benchmark.*

(ii) *If t and a are independent or positively associated, the activated corner $S^* = \{t \geq t_H, a \geq a_H\}$ satisfies $\text{CATE}_{S^*} \geq \text{ATE}$, with strict inequality unless $g(a) \omega(t)$ is almost surely constant.* (H1)

Proof. Equation (13) follows from (12) by iterated expectations: $E[\Delta] = \kappa E[g(a) \omega(t) E(D | t, a)] = \kappa \Delta \bar{x} E[g(a) \omega(t)]$. For (i), $\omega(t) < 1$ on $\{\delta(t) > 0\}$ and $\omega \leq 1$ everywhere (Lemma 1), so the expectation is strictly below one. For (ii), $h(t, a) = g(a) \omega(t)$ is increasing in each argument and $\mathbf{1}\{S^*\}$ is an increasing indicator. Independent random variables are associated (Esary, Proschan and Walkup, 1967), and positive association suffices by definition, so $\text{Cov}(h, \mathbf{1}\{S^*\}) \geq 0$, which gives $E[h | S^*] \geq E[h]$ and hence $\text{CATE}_{S^*} \geq \text{ATE}$ by (13) applied to S^* ; equality requires h almost surely constant. $\square$

Proposition 3 is H1 as a theorem. The average treatment effect is a predisposition-weighted mixture, and whenever trust and vigilance vary in the population, the population average sits strictly below the effect in the activated corner. A modest ATE and a large conditional effect are the same object seen at two levels of aggregation, which is the composition logic the empirical analysis tests. As in Canini, González and Prem (2026), these predictions motivate the empirical analysis; the model is a framework for interpretation, not a structure to be estimated.

S10 Pre-Registered Hypotheses and Deviations

S10.1 Pre-Registered Hypotheses (Verbatim)

The following hypotheses were pre-registered on the Open Science Framework prior to any analysis of the data. We reproduce them verbatim to ensure transparency regarding our confirmatory predictions.

Definitions:

- **Treatment manipulation:** The “Belief Correction” mechanism where participants estimate their own automation risk (0–100) and are then shown a dynamic visualization contrasting their subjective estimate against the objective risk score calculated by Frey and Osborne (2017), accompanied by an explanation of the expert methodology.
- **Policy response vector:** The five specific government interventions measured in Q96: (1) Expanding unemployment benefits and income subsidies, (2) Stricter federal regulations on AI deployment, (3) Funding retraining programs, (4) Strengthening labor protections (unionization), and (5) Limiting the number of jobs AI can automate.
- **“Policy Demand”:** The latent construct or composite score derived from the Policy Vector, representing the subject’s overall desire for government intervention to mitigate the workforce risks of AI.

The Conditional Activation Model suggests that looking for simple “Main Effects” is a theoretical error. Therefore, our hypotheses follow the logic of the funnel:

H1: The “Weak Main Effect” Hypothesis

The unconditional Average Treatment Effect (ATE) of the Treatment manipulation (Belief Correction) on Policy Demand will be small or statistically insignificant.

H2: The Trigger Hypothesis (Potential Energy) (Directional)

Treatment effects will be conditional on the Belief Gap. Individuals who discover they are at higher risk than they thought (Positive Shock) will show greater Policy Demand than those whose beliefs are confirmed or reassured.

H3: The Gate 1 Hypothesis (Trust Filter) (Directional)

The effect of the Belief Gap is moderated by Source Credibility.

- **High Trust:** The Gap significantly increases Policy Demand.
- **Low Trust:** The Gap is rendered inert; even a massive risk shock will not change Policy Demand if the source is rejected.

H4: The Gate 2 Hypothesis (Dispositional Filter) (Directional)

Among those who trust the source, the effect is further moderated by Trait Anxiety (a general psychological disposition).

- **High Trait Anxiety:** The objective risk signal resonates strongly, triggering a large increase in Policy Demand (Accuracy Motivation).
- **Low Trait Anxiety:** The signal triggers a baseline increase in Policy Demand, but significantly weaker than in high-anxiety individuals.

H5: The Conditional Activation Hypothesis (Directional)

This is the core test of the theory. The change in Policy Demand (Y) is conditional on surviving both gates:

- The strongest causal effects will be observed in the specific subset of participants who survive both filters: High-Trait-Anxiety Trusters who experience a Positive Belief Gap (i.e., this group will show significantly higher Policy Demand compared to the average participant or those with low Anxiety and low Belief Gap).

Note: The full pre-registration document, including the study design, sampling plan, analysis plan, and inference criteria, is available on the Open Science Framework. The OSF link is currently embargoed to preserve anonymity during peer review and will be made public upon acceptance.

S10.2 Deviations from the Registered Hypotheses

The paper reports four predictions (Section 2). They preserve the registered funnel: an aggregate prediction, a trust condition, a dose condition, and a dispositional condition. Table S15 maps each registered hypothesis to its reported counterpart and names the change. Two substantive deviations separate the registered and reported forms. Both postdate data collection. The registered forms above set the standard for judging the paper's tests.

Table S15: Crosswalk between registered and reported hypotheses.

Registered	Reported	Change
H1: Weak main effect	H1: Modest or null aggregate effect	None substantive.
H2: Trigger (Belief Gap)	H3: Risk-level gradient	Dose replaced: communicated risk level instead of the Belief Gap (first deviation).
H3: Gate 1 (trust filter on the gap)	H2: Trust gradient	Trust moderates the treatment effect, not the gap; tested by principal-score trust strata (second deviation).
H4: Gate 2 (trait anxiety among trusters)	H4: Anxiety amplification	None substantive.
H5: Conditional activation (trust $\times$ positive gap $\times$ high anxiety)	Joint prediction closing H4; corner test in Section 4	Dose as level, not gap (first deviation); corner set by a fixed top-tercile rule; the registration left the cutoff open.

The first deviation replaces the dose variable. The registered dose is the Belief Gap: the expert percentile minus the respondent's stated prior. The reported dose is the communicated risk level. The design forces the change. Only treated respondents state a prior, so the gap exists only in the treated arm. The gap-conditioned contrast therefore needs priors that were not measured among control respondents. Reconstructing them requires additional assumptions. The expert percentile needs no reconstruction. It is a fixed function of occupation, measured before treatment in both arms. Randomization alone supports contrasts conditioned on it. The paper keeps the gap as a manipulation check (Section 4): treated respondents perceived the discrepancy and reported updating in proportion to it. The paper therefore does not test registered H2 and H5 in that form.

The second deviation moves the trust test from the gap to the treatment effect. Registered H3 makes trust a moderator of the gap's effect within the treated arm. Reported H2 makes trust a moderator of the treated-versus-control contrast. The design cannot test the registered version. The survey measures trust after treatment and only among the treated. The naive alternative pits groups that were never balanced by randomization: high-rating treated respondents against all controls (Section 3.4). The paper instead estimates treatment effects within trust strata by principal-score weighting. The estimates are identified under randomization and principal ignorability (SI Section S1). Section 4 reports the breakdown values: how far the added assumption must fail before the estimates reach zero.

The pre-registration left one further choice open: the cutoff that defines the corner. Registered H5 places the strongest effect among high-anxiety trusters who experience a positive gap. The reported test places it in the

high-trust stratum where trait anxiety and the automation-risk percentile both fall in the pooled sample's top tercile. The rule estimates no parameters. The result replicates at two coarser cutoffs (Section 4), so the cutoff choice does not carry the finding. The reported test restates the registered conjunction on variables measured before treatment. The dose enters as level, not gap, per the first deviation.